第2章　公司金融实证研究方法

王义中
浙江大学经济学院金融系

- **合理性**
  - 变量度量是否合理
  - 变量选择是否合理
  - 描述性统计结果是否与既有文献类似
  - 回归系数大小是否超出经济学常识
  - 回归系数是否具有一定经济意义

- **一致性**
  - 理论分析与实证过程是否一致
  - 研究方法是否一致
  - 控制变量是否控制一致
  - 基准回归结果是否稳健

- 软件和数据
- 面板数据的处理：缩尾处理、描述性统计、相关系数
- 静态面板实证方法：固定效应、标准误聚类调整、控制变量、交互项、内生性问题、2SLS、两阶段选择模型、PSM、DID
- 动态面板实证方法（GMM）
- 其他实证方法（安慰剂检验、SUR）

软件和数据

- 计量软件的比较和选择
- 主流的计量软件有SPSS、SAS、Stata、R、Python等； 公司金融实证研究领域，Stata是目前最适合、应用最广泛的计量软件。

Features	SPSS	SAS	Stata	JMP (SAS)	R	Python (Pandas)
Learning curve	Gradual	Pretty steep	Gradual	Gradual	Pretty steep	Steep
User interface	Point-and-click	Programming	Programming/ point-and-click	Point-and-click	Programming	Programming
Data manipulation	Strong	Very strong	Strong	Strong	Very strong	Strong
Data analysis	Very strong	Very strong	Very strong	Strong	Very strong	Strong
Graphics	Good	Good	Very good	Very good	Excellent	Good
Cost	Expensive (perpetual, cost only with new version). Student disc.	Expensive (yearly renewal) Free student version, 2014	Affordable (perpetual, cost only with new version). Student disc.	Expensive (yearly renewal) Student disc.	Open source (free)	Open source (free)
Released	1968	1972	1985	1989	1995	2008

图片来源：https://dss.princeton.edu/training/StataTutorial.pdf

软件和数据

- **数据类型**
- 截面数据(cross-sectional data)：
  不同对象或主体在同一观测时间的数据集。
  例如2021年全国34个省级行政区的GDP数据。
- 时间序列数据（time-series data)：
  对某一变量多期观测的数据。
  例如2011-2021年中国的GDP数据。
- 面板数据（panel data / time-series cross-section data)：
  面板数据即为时间序列和截面混合的数据，即纵向为时间序列数据，横向为不同的变量。
  例如2011-2021年全国23个省份的GDP数据。
- 其他数据类型：
  离散数据、虚拟变量数据、社会调研数据、实验数据等。

软件和数据

- **数据来源**
  - 宏观数据来源

    中国统计年鉴、中国金融年鉴、IMF 各国宏观数据，包括 BOP、GFS、DOT、IFS

    OECD 、联合国、美联储、美国劳工部、商务部等网站、Wind 金融数据
  - 微观数据来源

    Wind、国泰安数据库

软件和数据

- 第一步，收集数据
- 第二步，转换成面板数据
- 第三步，缩尾处理（ winsorize ）
- 第四步，描述性统计
- 第五步，相关性分析
- 第六步，回归分析（工具变量、两阶段最小二乘、两阶段选择模型、PSM、DID、GMM、安慰剂检验、SUR）

软件和数据

- **第一步，收集数据**

  将字符转成数字：destring var1,replace; destring _all. Replace

  合并数据：merge 1:1 varlist using filename

- 注意：
- ① 绝对数与相对数：

  2008年GDP为30多万亿，2023年为121万亿。
  绝对数：为应对2008年金融危机，政府1万亿支出（实际是4万亿）；2023年为稳定经济，增发1万亿国债。
  相对数：1/121与1/30

- ② 单位统一：元、亿元；小数点与百分比
- ③ 公司匹配：不同时间下载数据
- ④ 变量度量要合理

软件和数据

- **第二步，转换成面板数据**

软件实现：

- 将字符转成数字： destring var1,replace; destring var1-var10, replace
- 合并数据：merge 1:1 varlist using filename
- 行列互换：reshape long a, i(gongsi) j(time)

  a为转换后变量，i 公司，j 生成时间
- 定义面板：tsset gongsi time

软件和数据

- **第二步，转换成面板数据**

- 平衡面板和非平衡面板：在平衡面板中，所有个体 i 的观测时间 t 是相同的，否则该面板就是一个非平衡面板。

- 软件实现：非平衡转换为平衡面板

  ssc install xtbalance,replace （需联网）

  qui xtset var1 var2

  xtdes

  xtbalance, range(2002 2008)

软件和数据

- **第三步，缩尾处理（winsorize）**
- 如果存在异常值，则需要进行缩尾处理（Winsorization）。进行99%分位和1%分位的缩尾处理，即变量取值高于99%分位数或者低于1%分位数时，分别以99%分位数和1%分位数代替。

- 先安装命令：ssc install winsor ； ssc install winsor2
- 软件实现：缩尾处理

  winsor var4, gen(var4_w) p(0.01)

  winsor2 var4 var5
- 公司层面的连续变量进行缩尾处理

软件和数据

- **第四步：描述性统计（summary statistics）**

- 离散程度指标
- 其一、方差与标准差。方差是每个数据值与全体数据平均数的差的平方的平均数。标准差是方差的算术平方根（即开方后的正根）。两者反映了一组数据的离散水平。

- 其二，四分位数。四分位数是指将所有数值由小到大排列并分成四等份后，处于 25%（下四分位数）和 75%（上四分位数）位置上的值。

软件和数据

- **第四步：描述性统计（summary statistics）**

- 集中趋势指标
- 其一，平均值。该指标容易被异常值和极端值所影响，可能难以反映数据集的真实特征。

- 其二，中位数。中位数是样本数据排序后最中间的数值当有极端值出现，无法用平均数很好地描述数据的集中情况，可以考虑使用中位数。

软件和数据

- **第四步：描述性统计（summary statistics）**

- **软件实现：描述性统计分析**
- tabstat var4 var5,stats(mean sd median max min count)
- mean（均值），count（非缺失观测值个数），

  sum（总和），max（最大值），min（最小值），range（最大值－最小值），sd（标准差），var（方差），median（中位数），skewness（偏度），kurtosis（斜度），

  p1(1％分位数，类似地有p5, p10, p25, p50, p75, p95, p99),

  iqr（interquantile range = p75 – p25）。
- tabstat var4,stats(mean), if var4==2001

  （计算所有公司2001年var4均值）

软件和数据

- 第四步：描述性统计正确示例

Summary statistics

	Citation-based network (1980–2006)			Text-based network (1996–2012)		
	Mean	Median	SD	Mean	Median	SD
A. Firm-specific variables						
log(sales)	4.858	4.966	1.964	4.022	4.024	2.004
Capx/assets	0.066	0.053	0.060	0.065	0.038	0.083
R&D/sales	0.208	0.052	0.668	0.147	0.003	0.585
PP&E/assets	0.252	0.223	0.167	0.261	0.189	0.225
Market-to-book	2.166	1.609	1.63	1.977	1.484	1.512
Book leverage	0.197	0.173	0.182	0.219	0.174	0.220
EBITDA/assets	0.106	0.135	0.193	0.087	0.118	0.204
WW	0.000	−0.015	1.000	0.000	0.022	1.000
SA	0.000	0.047	1.000	0.000	0.001	1.000
Delay	0.000	0.065	1.000	0.000	−0.090	1.000
Adjusted patents	0.682	0.328	0.868			
Adjusted citations	2.036	1.804	1.726			
Observations	17,230			50,865		
B. Competitor-pair averages						
MD	0.597	0.637	0.184			
Prod similarity				0.037	0.025	0.038
WW	0.003	−0.132	1.000	0.000	0.055	1.000
SA	0.000	−0.185	1.000	0.000	0.022	1.000
Delay	0.000	0.049	1.000	0.000	−0.070	1.000
No. of links per firm	20.596	13.000	36.407	56.812	28.000	64.37
Observations	354,871			2,889,742		

图片来源：Grieser 和 Liu（2019）

软件和数据

- **第五步，相关系数矩阵**

- 多重共线性问题。回归分析的基本假设之一就是解释变量之间相互独立，如果两个或多个解释变量之间出现了较强相关性，那么可能存在多重共线性问题.
- 相关系数矩阵，一般来说，相关系数小于0.4，无共线性（correlate 变量1　变量2 ….）
- 软件实现：

pwcorr var4 var5 ,sig star(.05)

corr var4 var5 或者correlate var4 var5

软件和数据

- 第五步，相关系数矩阵

- 方差膨胀因子（VIF）:根据经验法则，如果最大的方差膨胀因子VIF=Max(VIF$_1$,…, VIF$_n$) < 10，则表明不存在多重共线性问题。
- Stata软件实现

  reg var4 var5 var6

  vif

软件和数据

- **第六步，面板回归**

- 软件实现：reg var4 var5 var6 i.var1 i.var2,cluster(var1)
- 静态面板数据模型指解释变量中不包含被解释变量的滞后项（一般为一阶滞后），动态和静态模型在处理方法上存在差异。
- 最为常用的估计静态模型方法：面板OLS；

  估计动态模型方法：GMM（广义矩估计）+面板OLS 。
- 回归系数显著性；outreg2 using name.xls, tstat

静态面板实证方法

- reg var4 var5 var6 i.var2 i.var3,cluster(var1)
- est store model1
- reg var4 var5 var6 var7 i.var2 i.var3,cluster(var1)
- est store model2
- esttab model1 model2
- esttab model1 model2 using table.xls, replace
- set more off （关闭more项）

静态面板实证方法

Stata中用于估计Panel的常用命令

命令	模型
xtreg	Fixed-， between- and random-effects， and population-averaged linear models
xtregar	Fixed- and random-effects linear models with an AR(1) disturbance
xtgls	Panel-data models using GLS
xtpcse	OLS or Prais-Winsten models with panel-corrected standard errors
xtrc	Random coefficients models
xtivreg	Instrumental variables and two-stage least squares for panel-data models
xtivreg2	IV/2SLS, GMM and AC/HAC, LIML regression for panel data models (需要下载)
xtabond	Arellano-Bond linear dynamic panel-data estimator
xtdpdsys	Arellano-Bond/Blundell-Bond estimation
xtdpd	Linear dynamic panel-data estimation
xtabond2	Arellano-Bond system dynamic panel data estimator (需要下载)
xttobit	Random-effects tobit models
xtintreg	Random-effects interval data regression models
xtlogit	Fixed-effects， random-effects， population-averaged logit models
xtprobit	Random-effects and population-averaged probit models
xtcloglog	Random-effects and population-averaged cloglog models
xtpoisson	Fixed-effects， random-effects， population-averaged Poisson models
xtmixed	Multilevel mixed-effects linear regression
xtnbreg	Fixed-effects， random-effects， population-averaged negative binomial models
xtfrontier	Stochastic frontier models for panel-data
xthtylor	Hausman-Taylor estimator for error-components models
xtunitroot	Panel-data unit-root tests
xtwest	Westerlund error correction based panel cointegration tests
xtpmg	Pooled mean-group, mean-group, and dynamic fixed-effects models (需要下载)

静态面板实证方法

显著性

- 显著性水平：估计总体参数落在某一区间内，可能犯错误的概率或小概率事件发生的概率，用α表示。1-α 为置信度或置信水平，表明区间估计的可靠性。如果取α=0.01；0.05；010，说明正确的概率为99%；95%；90%。
- P值为拒绝域的面积或概率，为计量模型得到的精确的显著性水平，度量拒绝正确的原假设的概率。P值越大，则错误地拒绝原假设的概率越大；越小，则正确地拒绝的概率越大。
- 当P值< 0.01，则为***；当0.01< P值< 0.05.则为**；当0.05<P值<0.10，则为*。

静态面板实证方法

固定效应

- **公司固定效应（ Firm Fixed effect ）：** 不可观测的公司异质性，随个体变化但不随时间变化（例如，公司文化）
- **时间固定效应（ Time Fixed effect ）:**个体在时间上的异质性， 随时间变化但不随个体变化
- **行业固定效应（industry Fixed effect ）、国家、地区固定效应**
- 软件实现：

reg var4 var5 i.var1 i.var2

xtreg var4 var5 i.var2, fe(公司固定效应) ;

areg var4 var5 i.var2,absorb(var1)

静态面板实证方法

固定效应



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Debt enforcement, investment, and risk taking across countries☆

Giovanni Favara[a], Erwan Morellec[b,f,g,*], Enrique Schroth[c], Philip Valta[d,e,f]

Since firms may not be randomly assigned to the exposure groups, we control for the same time-varying observable firm characteristics as in the main specification (6). We also include firm fixed effects (α_i) to absorb time-invariant differences across firms and year fixed effects (δ_t) to control for time-varying factors common to all firms. Finally, we cluster standard errors at the firm level because firms' investment and risk choices may be correlated over time.

- **公司固定效应：**

  吸收不同公司间的时间固定不变的差异

- **时间固定效应：**

  控制对所有公司而言时间可变因素

静态面板实证方法

标准误的聚类调整

- standard errors corrected for clustering
- when you think of random changes in the crime rates in a particular county, some of the factors causing crime to change just operate within that county, while others operate statewide. So it is necesssary to make a clustering correction to the standard errors in the data.
- 面板数据pool的特点是除影响特定地区的聚类因素外，还有共同的因素影响聚类。另外，不进行标准误聚类调整会高估显著性

静态面板实证方法

标准误的聚类调整

- 软件实现
- regress var3 var4,cluster(var1)；xtreg var3 var4,fe cluster(var1) 一维
- 二维标准误聚类调整（cluster2）
  http://www.kellogg.northwestern.edu/faculty/petersen/htm/papers/se/se_programming.htm

Cluser2 var3 var4,fcluster(var1) tcluster(var2)

reghdfe var3 var4,a(var1 var2) cluster(var1 var2)

(含固定效应)

静态面板实证方法

控制变量

- 被解释变量与解释变量，因变量与自变量，控制变量与非控制变量（主要解释变量或主要自变量）
- 控制变量（物理学）是指那些除了实验因素以外的所有影响实验结果的变量，这些变量不是本实验所要研究的变量，所以又称非实验因素。只有将自变量以外一切能引起因变量变化的变量控制好，才能弄清实验中的因果关系。
- **Ownership structure and the cost of corporate borrowing（JFE，2011）**

2.4. Control variables

To assess the impact of the control-ownership wedge on loan spreads, we control for other factors that might affect loan pricing. These factors include borrower characteristics, loan characteristics, macroeconomic factors, as well as borrower industry and year fixed effects. The reasons for including these control variables in loan spread models are relatively well-known, so we provide only a brief discussion below. Detailed definitions for all the variables used in the paper are provided in Table 1.

静态面板实证方法

控制变量回归系数的显著性和大小

Table 2
The effect of CEO tenure on CDS spread

	(1) Years [0,9]	(2) Years [0, 2]	(3) Years [0, 2], Before 2008	(4) Years [0, 2]	(5) Years [0, 2]
	5-year CDS spread	5-year CDS spread	5-year CDS spread	3-year CDS spread	1-year CDS spread
Tenure (years 0-2)	−0.031*** (0.011)				
Tenure (years 3-5)	−0.018 (0.021)				
Tenure (years 6-9)	−0.003 (0.011)				
Tenure (in days)		−0.031** (0.015)	−0.044*** (0.016)	−0.058*** (0.018)	−0.080*** (0.020)
Stock price	−1.369*** (0.188)	−0.962*** (0.244)	−0.892*** (0.247)	−0.943*** (0.280)	−0.953*** (0.299)
Recovery rate	−12.512*** (1.226)	−9.587*** (1.442)	−8.626*** (1.531)	−15.655*** (1.967)	−20.063*** (2.176)
Credit spread	0.450*** (0.040)	0.406*** (0.065)	0.409*** (0.082)	0.483*** (0.080)	0.624*** (0.096)
Term spread	−0.017 (0.029)	−0.042 (0.043)	−0.005 (0.048)	−0.052 (0.051)	−0.008 (0.057)
VIX	1.655*** (0.138)	2.225*** (0.248)	2.584*** (0.331)	2.182*** (0.290)	1.996*** (0.318)
Log(assets)	−20.458* (12.339)	76.013** (32.443)	52.854 (33.414)	97.048** (38.390)	125.716*** (41.739)
Leverage	210.559*** (46.624)	82.981 (59.360)	40.336 (59.425)	128.798* (73.450)	100.586 (78.873)
M/B	9.251 (6.317)	14.190 (12.748)	16.734 (13.566)	23.549 (17.416)	31.336* (18.916)
ROA	−402.824*** (70.904)	−486.699*** (115.805)	−574.895*** (137.286)	−570.857*** (136.150)	−584.902*** (146.363)
Tangibility	17.383 (76.118)	54.684 (149.272)	−233.555 (149.465)	228.103 (175.092)	259.161 (181.409)
CF volatility	15.802*** (4.426)	8.184 (7.133)	6.276 (7.515)	7.929 (8.204)	8.659 (9.709)
Payout ratio	−23.535*** (5.095)	−7.458 (4.958)	−8.204 (6.193)	−13.331** (5.901)	−14.017** (6.309)
Firm-CEO,Yr FE	x	x	x	x	x
Observations	757,434	301,290	239,516	274,324	266,325
Adj. R^2	0.755	0.840	0.829	0.807	0.749

Variable	log(RevPAR) (1)	log(RevPAR) (2)	log(RevPAR) (3)	log(RevPAR) (4)
Cohort Effect$_{ih}$	−0.0040*** (0.0015)		0.0021 (0.0017)	
Entrants$_{ich}$		−0.0068*** (0.0015)	−0.0070*** (0.0016)	−0.0070*** (0.0016)
Cohort Effect$_{ih}$ (levels)				0.0080 (0.0062)
log(Income)	−0.0090 (0.0455)	−0.0041 (0.0436)	−0.0043 (0.0435)	−0.0043 (0.0435)
Unemployment	−0.0184*** (0.0040)	−0.0191*** (0.0039)	−0.0192*** (0.0039)	−0.0192*** (0.0039)
log(Popul.)	−0.1051*** (0.0361)	−0.1011*** (0.0357)	−0.1012*** (0.0357)	−0.1012*** (0.0357)
log (Housing Prices)	0.2294*** (0.0264)	0.2238*** (0.0253)	0.2238*** (0.0253)	0.2238*** (0.0253)
Hotels in County	−0.0002** (0.0001)	−0.0001 (0.0001)	−0.0001 (0.0001)	−0.0001 (0.0001)
log(AE&R estab.)	0.0751** (0.0329)	0.0661** (0.0320)	0.0659** (0.0320)	0.0659** (0.0320)
log(F&B estab.)	0.0839** (0.0345)	0.0873*** (0.0338)	0.0875*** (0.0338)	0.0875*** (0.0338)
log(Acc. estab.)	0.0053 (0.0153)	0.0111 (0.0151)	0.0113 (0.0151)	0.0113 (0.0151)
log(Rooms)	−0.0549*** (0.0127)	−0.0554*** (0.0124)	−0.0554*** (0.0124)	−0.0554*** (0.0124)
Age	−0.0132*** (0.0011)	−0.0135*** (0.0011)	−0.0136*** (0.0011)	−0.0135*** (0.0011)
Age^2	0.0002*** (0.0000)	0.0002*** (0.0000)	0.0002*** (0.0000)	0.0002*** (0.0000)
Location type fixed effects	Yes	Yes	Yes	Yes
Org. form fixed effects	Yes	Yes	Yes	Yes
Brand fixed effects	Yes	Yes	Yes	Yes
Segment-year fixed effects	Yes	Yes	Yes	Yes
County clustering	Yes	Yes	Yes	Yes
R^2	0.6615	0.6636	0.6636	0.6636
N	219,849	219,849	219,849	219,849

- JF（2016）
- 不用过于关注控制变量的显著性；重装饰

静态面板实证方法

交互项和边际效应

- **Liquidity needs and vulnerability to financial underdevelopment（JFE，2006）**

The hypothesis that financial development has a relatively greater impact on the volatility of industries with high liquidity needs can be tested by estimating the parameters of the following specification:

$$V_{i,k} = \alpha_k + \beta_i + \gamma L_i x F_k + X_{i,k}\delta + \varepsilon_{i,k}, \quad (1)$$

where $V_{i,k}$ is the volatility of industry i in country k, α_k is a country-specific effect, β_i is an industry-specific effect, L_i is the measure of liquidity needs of the industry, F_k is the measure of financial development of the country, $X_{i,k}$ is the set of additional determinants of sector volatility, ε_{ij} is a random error, and δ and γ are parameters to be estimated. The parameter of interest is γ, or the effect of the interaction of financial development and liquidity needs on volatility. If financial development leads to a relatively greater reduction in the volatility of sectors with high liquidity needs, then γ should be negative and economically significant.

静态面板实证方法

交互项和边际效应

- **Financial Innovation： The Bright and the Dark Sides（2012）**

Bekaert et al. (2007) show that exogenous growth opportunities predict subsequent GDP growth and more so in countries with liberalized capital accounts, banking systems and equity markets. We build on this work and gauge the relationship between financial

$$Growth_{i,t} = \beta_1 GGO_MA_{i,t} + \beta_2 FI_{i,t} + \beta_3 GGO_MA_{i,t} * FI_{i,t} + \gamma X_{i,t} + \varepsilon_{i,t}, \quad (1)$$

where $Growth_{i,t}$ is the five-year moving average annual real GDP per capita or real capital growth in country i and period t; GGO_MA is a measure of global growth opportunities and FI is one of our indicators of financial innovation. Following Bekaert et al. (2007), we use

静态面板实证方法

交互项和边际效应

- **The role of relative price volatility in the efficiency of investment allocation, Journal of International Money and Finance 33 (2013) 1–18**

Based on the framework subsumed in Equation (4) of the previous section, we estimate regressions of the following type:

$$\frac{I_{ijt}}{\overline{I_{jt}}} = \gamma_1 \frac{\mathrm{TFP}_{ijt-1}}{\overline{\mathrm{TFP}_{jt-1}}} + \gamma_2 \frac{\mathrm{TFP}_{ijt-1}}{\overline{\mathrm{TFP}_{jt-1}}} \cdot \sigma_{jt-1} + \mu_{jt} + \nu_{it} + \varepsilon_{ijt} \tag{5}$$

where i,j and t denote a sector, a country, and year respectively, I_{ijt} is investment in sector i in country j at time t, $\overline{I_{jt}}$ is total investment in country j at time t ($\overline{I_{jt}} = \sum_i I_{ijt}$), and TFP_{ijt} is a measure of total factor productivity of sector i in country j at time t. $\overline{\mathrm{TFP}_{jt}}$ is the sum of total factor productivities of all sectors in country j at time t. σ measures relative price volatility. Finally, μ and ν are country-time and industry-time fixed effects. These dummy variables control for all possible observable and unobservable components that vary at a country-time and sector-time level, respectively, reducing the need for additional control variables.

These results are not only statistically significant; their economic impact is also relevant. To see this, and to visualize how the impacts differ across groups of countries, Fig. 2 plots the marginal impact of changes in the TFP ratio on investment shares for different levels of volatility and for each group of countries. The marginal impact is computed as:

$$\frac{\partial \frac{I_{ijt}}{\overline{I_{jt}}}}{\partial \frac{\mathrm{TFP}_{ijt-1}}{\overline{\mathrm{TFP}_{jt-1}}}} = \gamma_1^k + \gamma_2^k \sigma_{jt-1} \tag{7}$$

静态面板实证方法

- **理论机制/作用机制/作用渠道与异质性分析**
- 如果x1影响x2，然而影响y，则x2是影响机制，但有个前提：x2是连续变量，而不是虚拟变量。如果x2是虚拟变量，则x2只是既定事实，而不受x1影响。例如，x2为企业性质，国有企业和民营企业，则x2是既定事实而不是作用机制。作用机制一定是受到x1影响且会变化的变量。
- 作用机制回归方程：y=x1+x2+x1*x2

静态面板实证方法

经济意义与统计意义

- 自变量为连续变量（一个标准差的增加——样本均值）

Table 2
Within-firm evidence on loan vs. bond choice, benchmark regressions.

Dependent variable:	D_{it}=1 if a firm i receives a loan and D_{it}=0 if a firm issues a bond in quarter t					
Mean, D_{it}:	0.135 (1)	0.135 (2)	0.148 (3)	0.148 (4)	0.135 (5)	0.135 (6)
Tightening in lending standards	−0.059**	–	–	–	–	–
	(0.026)					
Aggregate lending growth	–	0.260***	–	–	–	–
		(0.064)				
Non-performing loans	–	–	−1.030***	–	–	–
			(0.160)			
Loan allowances	–	–	–	−5.122***	–	–
				(0.854)		
Bank stock-index	–	–	–	–	0.108***	–
					(0.021)	
Monetary policy	–	–	–	–	–	−1.617***
						(0.571)
Log (Assets)	0.039*					
	(0.021)					
Log (PP&E)	−0.017					
	(0.019)					
ROA	−0.375					
	(0.231)					
Market-to-book	−0.026**					
	(0.013)					
Lagged return	−0.035***					
	(0.012)					
Leverage	−0.037					
	(0.049)					
Dividend payer	−0.096***					
	(0.024)					
Firm fixed effects (obs.)	Yes (2288)	Yes (2288)	Yes (2158)	Yes (2158)	Yes (2288)	Yes (2288)
Clusters (quarter)	83	83	68	68	83	83
R-squared	0.39	0.39	0.40	0.40	0.39	0.39
Observations	9670	9670	8156	8156	9670	9670

Table 2 presents our first main result: Eq. (1) is estimated using ordinary least squares (OLS), with errors clustered by quarter. All six time-series variables indicate a strong pro-cyclical pattern in the debt financing mix for the firms in our sample. In column 1, the cyclical variable is the net fraction of loan officers reporting tightening credit standards, predicted to be negatively correlated with banks' willingness to lend. Indeed, the coefficient is negative and significant. The coefficient point estimate, −0.059, implies that a one standard deviation increase in lending standards is predicted to decrease the probability that a firm gets a loan, conditional on receiving debt financing, by 1.4 percentage points (or, equivalently, that the fraction of external debt financings that is made up of bank loans will be lower by 1.4 percentage points). This is a 10% increase over the sample mean. In other words, firms appear to substitute bonds for bank loans at times when lending standards are tight.

静态面板实证方法

经济意义与统计意义

- 自变量为虚拟变量

This table reports the effect of an IPO on the fundamental nature of research. In columns (1) to (3) the dependent variable is the average scaled originality in the five years after the IPO filing, and in columns (4) to (6) it is average scaled generality. *IPO* is a dummy variable equal to one if a firm completes the IPO filing, and zero otherwise. *NASDAQ Returns* is the two-month NASDAQ returns calculated from the IPO filing date. In columns (1) to (3) I control for the prefiling average scaled originality, and in columns (4) to (6) I control for the corresponding generality measure. Additional control variables are: prefiling average scaled citations, prefiling average scaled patents per year, *VC Backed, Pioneer, Early Follower*, and the three-month NASDAQ returns before the IPO filing. All variables are defined in the Appendix. The model is estimated using OLS in columns (1), (2), (4), and (5) and using 2SLS in columns (3) and (6). *Magnitude* is the ratio of *IPO* coefficient to the prefiling average of scaled originality or scaled generality. Robust standard errors are reported in parentheses. *, **, and *** indicate that the coefficient is statistically significant at the 10%, 5%, and 1% level, respectively.

Dependent Variable Model	(1) Scaled Originality OLS	(2) Scaled Originality OLS	(3) Scaled Originality 2SLS	(4) Scaled Generality OLS	(5) Scaled Generality OLS	(6) Scaled Generality 2SLS
IPO	−0.006		−0.137**	−0.001		−0.087
	(0.010)		(0.068)	(0.016)		(0.092)
NASDAQ returns		−0.081**			−0.050	
Magnitude	−0.10%					
Observations	1,079					
R^2	0.231					
Filing year FE	Yes					
Industry FE	Yes					
Control variables	Yes	Yes	Yes	Yes	Yes	Yes

the IPO significantly declines as the IPO coefficient equals −0.137, reflecting a decline of 13% ($=-\frac{0.13}{1.06}$, where 1.06 is the average scaled originality in the pre-event years). These findings suggest that issuers who remain private produce patents that rely on a broader set of technologies. In columns (4) to (6), I repeat

静态面板实证方法

经济意义与统计意义

- 自变量为交互项的回归模型设定

$$\frac{I_{i,t}}{A_{i,t-1}} = \alpha_i + \alpha_t + \beta_1 q_{i,t-1} + \beta_2 \frac{CF_{i,t}}{A_{i,t-1}} + \beta_3 q_{i,t-1} \times E_t + \beta_4 \frac{CF_{i,t}}{A_{i,t-1}} \times E_t + \beta_5 q_{i,t-1} \times S_t + \beta_6 \frac{CF_{i,t}}{A_{i,t-1}} \times S_t + \varepsilon_{i,t}.$$

To evaluate the economic significance of the effects, consider the sixth regression, which includes the expansion interactions, along with the Baker and Wurgler sentiment interactions. In this regression, the q-expansion interaction coefficient is 0.133 (t-statistic = 6.53), while the cash flow expansion interaction coefficient is −0.401 (t-statistic = 4.99). The q-sentiment interaction coefficient is 0.212 (t-statistic = 4.14), while the cash flow–sentiment interaction coefficient is −0.648 (t-statistic = 2.45). Assume that the rank of the Baker and Wurgler sentiment index takes on its mean value of 0.489. In this case, the overall q coefficient is $0.158 + 0.212 \times 0.489 + 0.133 = 0.395$ in expansions and $0.158 + 0.212 \times 0.489 = 0.262$, or 33% lower, in recessions. The overall cash flow coefficient is 0.362 in expansions and 0.763, or 108% higher, in recessions. Hence, the effects of a recession on investment sensitivities are not only statistically significant, but also economically important.

- McLean 和 Zhao（2014）

静态面板实证方法

经济意义与统计意义

- 自变量为交互项的回归结果示例

因变量为投资	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$q(t-1)$	0.403***	0.292***	0.315***	0.281***	0.322***	0.158***	0.220***	0.163***	0.250***
	(14.03)	(14.88)	(19.36)	(8.38)	(9.67)	(4.82)	(6.36)	(3.86)	(6.83)
Cash flow	0.340***	0.693***	0.618***	0.706***	0.660***	1.080***	0.980***	1.019***	0.872***
	(3.99)	(8.85)	(7.81)	(4.50)	(4.42)	(6.59)	(6.20)	(5.92)	(5.18)
Expansion × $q(t-1)$		0.125***				0.133***	0.120***		
		(4.96)				(6.53)	(5.57)		
Expansion × cash flow		−0.392***				−0.401***	−0.369***		
		(4.42)				(4.99)	(5.01)		
Indus.Prod × $q(t-1)$			0.111***					0.129***	0.103***
			(3.71)					(4.07)	(3.28)
Indus.Prod. × cash flow			−0.332***					−0.349***	−0.290***
			(3.44)					(3.71)	(3.03)
Sentiment × $q(t-1)$				0.204***		0.212***		0.231***	
				(3.22)		(4.14)		(4.32)	
Sentiment × cash flow				−0.628**		−0.648**		−0.660**	
				(1.99)		(2.45)		(2.47)	
CSI × $q(t-1)$					0.131**		0.123**		0.114**
					(2.13)		(2.32)		(2.10)
CSI ×cash flow					−0.516**		−0.497**		−0.468**
					(2.19)		(2.31)		(2.15)
Observations	147,271	146,823	146,823	147,271	146,823	146,823	146,823	146,823	146,823
R^2 (within)	15.43	15.82	15.87	15.93	15.81	16.35	16.16	16.46	16.15

静态面板实证方法

稳健性检验

- Jayaraman, S., and Milbourn T. 2012. "The Role of Stock Liquidity in Executive Compensation." *The Accounting Review* 87 (2) (March): 537-563.
- 回归系数没有出现跳跃性变化；回归系数显著性没有出现明显变化；回归系数的方向没有变化
- Robustness Tests
  - Using alternative measures of PPS
  - Alternative measures of stock liquidity
  - Using CEO cash compensation
  - Controlling for the influence of other unobservable performance measures

静态面板实证方法

稳健性检验

3. 稳健性检验[①]

（1）计量方法变化。首先，采用 Logit 模型对方程（1）进行回归，四个信贷供给周期指标的系数都是显著为正，将四个信贷供给周期变量系数标准化后，其系数分别为：0.142、0.133、0.224 和 0.051[②]。无论是系数符号还是系数大小排序都与基准回归相符合。其次，因为存在货币政策变化的“银行信贷渠道”（Bernanke & Blinder，1988），即货币政策变化会导致银行信贷供给的变化，所以我们采取滞后 1 期的 M2 增长率作为工具变量（用 M2 增长率衡量货币政策变化），采用两阶段最小二乘法（2SLS）对模型进行估计，除以银行业景气指数变化率作为信贷供给周期指标时显著性有所降低外，其他指标考虑内生性问题后得到的经验结果与基准回归结果相一致。

（2）样本量变化。首先，放松两个季度连续的样本选择标准，采用最初的大样本（6052 个观测值）来估计方程（1），得到的经验结果与基准回归相符。同时，对大样本采用 Logit 模型进行分析，结论同样与基准回归相一致。其次，将删除的同一季度有两个不同融资目的的观测值加入到样本中，并赋予其近似融资目的:如果一个公司在同一个季度内有两种融资目的，则先根据不同目的贷款的笔数判断，如果流动性资金需求的贷款数量大于长期资金需求，则判定该公司在该季度的融资目的为流动性需求，反之则是长期资金需求；如果两种贷款的数量相等，则根据其贷款金额判断，该公司在该季度内流动性需求贷款金额总和大于长期资金需求的贷款，则认为该公司的融资目的为流动性资金需求，反之则为长期资金需求。接着，按照原来的方程（1）对新的样本进行估计，结论与前文一致。

静态面板实证方法

稳健性检验

（五）稳健性检验②

1. 内生性问题。借鉴 Wintoki et al.（2012），我们采用系统广义矩估计（GMM）方法对基准回归方程进行检验。广义矩估计主要适用于动态面板估计，能够减轻模型中可能存在的因变量与自变量之间相互影响、遗漏变量等所产生的内生性问题（Wintoki et al.，2012）。估计结果与表 3 基本一致。另外，鉴于中国宏观经济不确定性与美国宏观经济之间较强的相关性，我们采用滞后 1 期美国宏观经济不确定性作为工具变量，运用二阶段最小二乘法（2SLS）进行估计。经验结果显示，各变量的回归系数没有出现明显变化。

2. 其他稳健性检验

（1）考虑影子银行问题。上市公司除通过银行贷款、发行股票和债券等方式获得资金来源外，还可能通过其他方式获得资金。影子银行是从事到期、信贷和流动性转换等业务的金融中介，但它

（2）考虑一年内到期的长期借款。一年内到期的长期借款会转入“一年内到期的非流动负债”科目中。如果不考虑一年内到期的长期借款额，可能会低估公司的长期资金需求规模。为此，我们将“一年内到期的非流动负债”（包括一年内到期的长期借款、应付债券和长期应付款）加入到长期资金需求中，并重新进行经验分析，得到的结果与基准回归结果（表 3）相一致。③

（3）调整资金需求比例。除了检验内生性问题和检验影响的异质性之外，我们还做了如下稳健性检验：按照流动性资金需求占 30% 和长期资金需求占 70% 的比例，重新调整流动性资金需求和长期资金需求变量；宏观经济不确定性用工业增加值的条件方差进行度量。我们发现，各变量的回归系数没有出现明显变化，回归结果是稳健的。

静态面板实证方法

内生性与工具变量

- 内生性：模型中的一个或多个解释变量与随机扰动项相关。内生变量与外生变量。
- 例1：教育水平（X1）和能力（X2）影响工资，但能力不可观测，找不到可以度量的变量。回归：$Y=X1+\varepsilon$。

  问题：误差项中包含着能力，且教育水平与能力相关，所以出现教育的内生性问题。
- 例2：经济增长=金融发展+?
- 例3：创新=高房价+？；
- 例4：2008年雪灾方便面高价=高需求+?

静态面板实证方法

发现内生性问题与解决方法？

- 核心解释变量是否与没有控制的因素有联系，或核心解释变量不是严格外生变量？如果遗漏的变量与其他解释变量不相关，一般不会造成问题。否则，就会造成解释变量与残差项相关，从而引起内生性问题。

  方法：工具变量、2SLS

- 解释变量与被解释变量是否存在因果联系。

  方法：解释变量滞后一期；PSM; DID

- 解释变量的度量上存在误差，使其与真实值之间存在偏差，这种偏差可能会成为回归误差（regression error）的一部分，导致内生性问题。

  方法：增加变量不同的度量方式; 样本选择偏误 两阶段选择模型。

静态面板实证方法

方法 1　两阶段最小二乘

- 2SLS主要解决自变量与误差项相关的内生性
- 工具变量instrumental variable：某一个变量与模型中随机解释变量（自变量）高度相关，但却不与随机误差项相关，估计过程中被作为工具使用，以替代模型中与误差项相关的随机解释变量的变量，称为工具变量
- 作为工具变量，必须满足四个条件：与所替的随机解释变量高度相关；与随机误差项不相关；与模型中其他解释变量不相关；同一模型中需要引入多个工具变量时，这些工具变量之间不相关
- 例：农民收入=储蓄+？。但储蓄也会影响收入。用地区降雨量（天气）作为农民收入的工具变量。

静态面板实证方法

方法 1　两阶段最小二乘

- 最简单的工具变量：

  （1）用行业/地区平均值作为某一个公司对应变量的IV进行内生性检验。

  （2）用内生变量的自身滞后1期（或多期）作为工具变量。

- Lin, C., P. Lin, F. Song, C. Li, 2011, "Managerial Incentives, CEO Characteristics and Corporate Innovation in China's Private Sector", *Journal of Comparative Economics* 39, 176-190.

- Lin, C., Y. Ma, P. Malatesta, Y. Xuan, 2012, "Corporate Ownership Structure and Bank Loan Syndicate Structure", *Journal of Financial Economics* 104, 1-22

- 李春涛和宋敏.中国制造业企业的创新活动:所有制和CEO激励的作用.《经济研究》2010年第5期

静态面板实证方法

方法 1　两阶段最小二乘

使用借款企业所属行业的期初行业平均控制权-所有权偏离度，以及该行业期初平均现金流权，分别作为对应工具变量。

3.2.2. Instrumental variable estimation

Our second approach to address the endogeneity issue is to employ instrumental variables estimation. We instrument for each borrowing firm's control-ownership wedge and cash-flow rights using the initial industry average wedge and the initial industry average cash-flow rights in the borrower's industry as respective instruments (Laeven and Levine, 2009).[17] The initial industry averages are calculated in the year prior to the start of our sample and are country-specific. As Laeven and Levine (2009) note, an individual firm's ownership structure is correlated with its industry average ownership structure. However, it is unlikely that the average industry ownership structure directly affects the syndicate structure of any particular loan of the borrowing firm except through the borrower's ownership structure. In the first-stage regression (untabulated), the

静态面板实证方法

方法 1　两阶段最小二乘

- 例子1：教育水平（X1）和能力（X2）影响工资。能力不可观测，且没有代理变量IQ。进行回归：Y=X1+C
- 问题：误差项中包含着IQ（能力），且教育水平与能力相关，出现教育的内生性问题。
- 一个好的工具变量（IV）与教育水平高度相关，并且与误差项不相关。IQ作为工具变量？不，它同时与教育和误差项相关。
- 文献中使用的IV:母亲教育水平；兄弟姐妹数量：兄弟姐妹越多，平均受教育程度越低。IV需要与能力不相关。

静态面板实证方法

方法 1　两阶段最小二乘

results. We will have as many first-stage regressions as there are endogenous explanatory variables, and you need at least one significant coefficient on the instruments in each reduced form regression for the model to be well identified. This is a necessary, not sufficient condition, however. To see this, consider the model

外生变量

$$y = \beta_1 + \beta_2 x_2 + \beta_2 x_3 + \beta_3 x_4 + u,$$

where x_3 and x_4 are endogenous. We need at least two instruments, say z_1 and z_2, and these enter in the reduced form equations for x_3 and x_4:

$$x_3 = \pi_1 + \pi_2 x_2 + \pi_3 z_1 + \pi_4 z_2 + \varepsilon_1,$$

$$x_4 = \gamma_1 + \gamma_2 x_2 + \gamma_3 z_1 + \gamma_4 z_2 + \varepsilon_2.$$

- If $\pi_3 = 0, \pi_4 \neq 0, \gamma_3 = 0, \gamma_4 \neq 0$, the structural equation is **not** identified, because the instrument z_1 is irrelevant in both equations - hence, effectively we only have one instrument.
- If $\pi_3 = 0, \pi_4 \neq 0, \gamma_3 \neq 0, \gamma_4 = 0$, the structural equation **is** identified, because the instrument z_1 is relevant in the equation determining x_4, while the z_2 is relevant for x_3.

静态面板实证方法

方法 1　两阶段最小二乘

- 疑问：第1阶段，内生变量与工具变量回归中，为什么要放入外生变量x2?

  情况1：不放入外生变量，即x3=z1+z2作为第1阶段；

  情况2：放入外生变量，即x3=x2+z1+z2作为第1阶段；

  情况3：放入其他外生变量，即x3=x5+x6+x2+z1+z2

- 以上三种情况在文献中都出现过，不会改变第2阶段的回归系数方向，只会改变回归系数大小。但x2放入第1阶段时较为合理的，因为x2作为模型中的严格外生变量（非内生变量）是影响其他变量的变量，而不能受其他变量的影响。否则， x2需要作为因变量而另设一个回归方程。

静态面板实证方法

方法 1　两阶段最小二乘

From a practical point of view, you will almost certainly notice if identification fails. If your model is literally not identified, because you have too few instruments or because the instruments are collinear, then Stata will report this and stop. If your instruments are very weakly correlated with the endogenous explanatory variables, the coefficients on the instruments in the first stage may be insignificant, and the 2SLS standard errors very large - correctly telling you that you are not learning anything from the current model.

- 严重识别失败（工具不够、完全共线）：Stata停止运行；
- 弱工具问题：不会报错，但一阶段系数不显著、2SLS标准误膨胀，估计结果不可信。

静态面板实证方法

方法 1　两阶段最小二乘

- 软件实现
- 两阶段最小二乘法（2SLS）

第一阶段：reg var5（内生变量） var1 var2（工具变量）

predict v_w （存储，预测回归的拟合值）

第二阶段：reg var4 v_w

缺陷：无法判断工具变量的好坏

静态面板实证方法

方法 1　两阶段最小二乘

- 软件实现 另一种方法

- ivreg2 因变量y 自变量x2 (自变量x1=工具变量z3 工具z4 工具z5) , endog(x1)

例子：ivreg2 var4 (var5=var1 var2),endog(var5)

需要下载命令findit ivreg2（找到SJ-7-4）

- Stock-Yogo weak ID test：原假设是所选工具变量是弱工具变量，当Cragg-Donald统计量大于Stock-Yogo weak ID test critical value在10%的水平时，拒绝原假设，即不存在弱工具变量的问题。

静态面板实证方法

方法 1　两阶段最小二乘

- **Jayaraman, S., and Milbourn T. 2012. "The Role of Stock Liquidity in Executive Compensation." *The Accounting Review* 87 (2) (March): 537-563.**

We explore the role of stock liquidity in influencing features of managerial compensation contracts. We find that the proportion of cash-based compensation to total compensation is decreasing for firms with greater stock liquidity. Further, the reliance placed on stock prices relative to earnings as alternative benchmarks for performance-based pay is higher for more liquid firms. These results are consistent with optimal contracting theories that suggest that greater stock liquidity improves the informational content of stock prices and consequently their use in compensation contracts. We also use regressions that rely on changes in both dependent and independent variables to provide evidence on the relation between declining cash-based compensation and increasing stock liquidity over time. Finally, we use an exogenous shock to stock liquidity to provide evidence on the causality between stock liquidity and features of executive compensation contracts. Overall, our study offers stock liquidity as a possible contributor of why the composition of annual compensation and the weights on stock prices versus earnings have been changing over time.

静态面板实证方法

内生性与工具变量

- 股票的流动性对经理人的激励合约的设计会产生影响。具体的说，当流动性提高的时候，经理人会更加倾向于选择cash-based compensation比重更小的激励合同，因为此时股票变现成本较低
- 被解释变量分别是cash-based compensation占total compensation的比重，自变量是流动性

内生性来源：

1. 反向因果：不只是流动性影响薪酬结构；高管激励合约设计本身，也会反过来影响公司股票流动性。

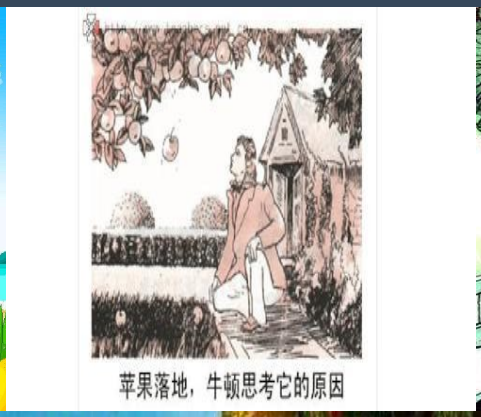
   例：大量股权激励会增加股票交易，改变流动性。

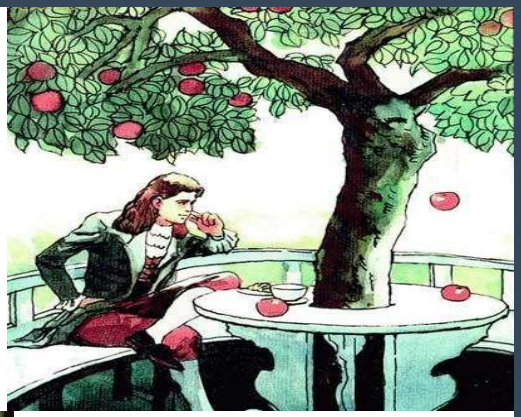
2. 遗漏变量：公司治理、成长性、风险、行业特征，会同时影响流动性和薪酬设计。

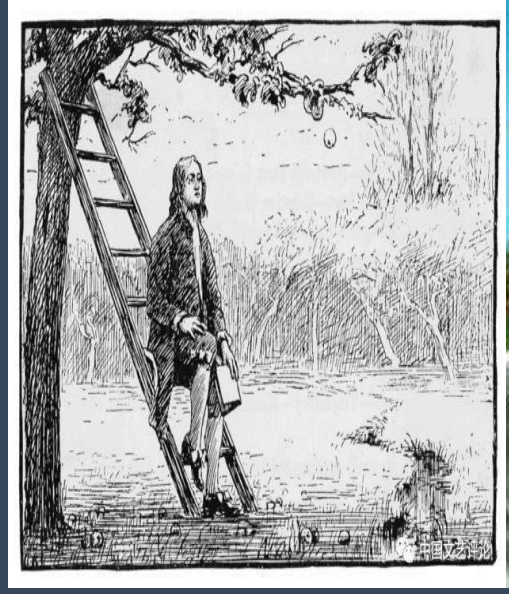
静态面板实证方法

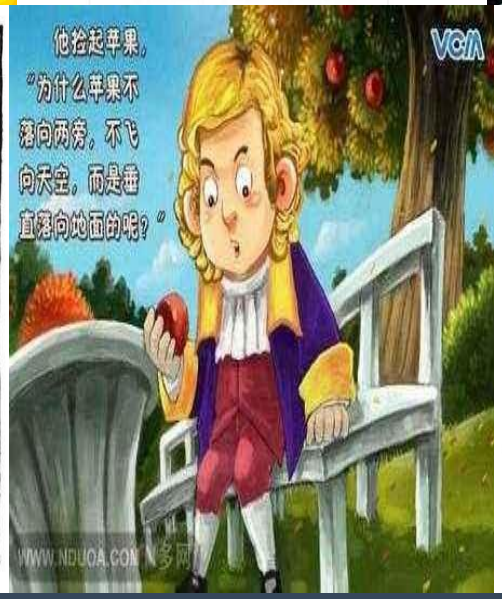
内生性与工具变量











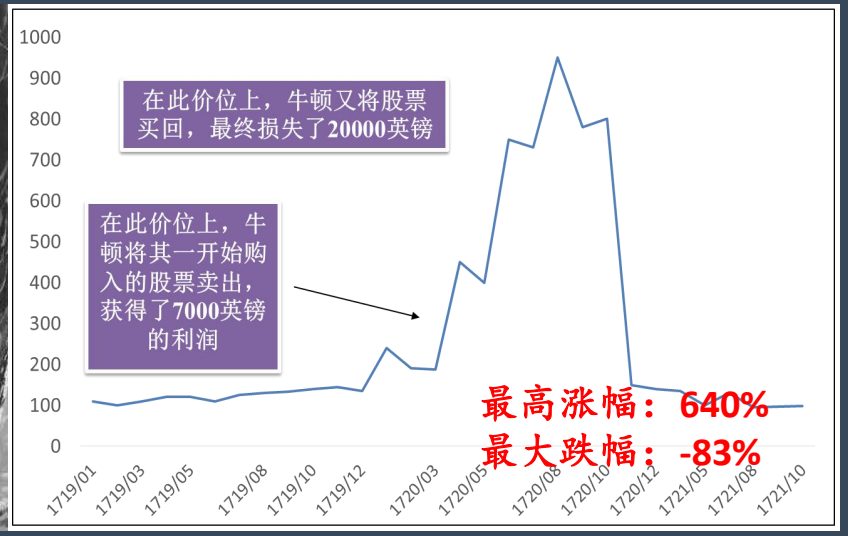


静态面板实证方法

内生性与工具变量



我可以计算天体运动，却无法计算...



南海贸易公司股票价格的涨落

静态面板实证方法

内生性与工具变量

- **Relation between cash-based compensation and stock liquidity**

H1a: Firms with greater stock liquidity are associated with a lower proportion of cash-based compensation to total compensation.

H1b: Firms with greater stock liquidity are associated with a higher proportion of cash-based compensation to total compensation.

$$Cashcomp = \alpha_0 + \alpha_1 Liq + \alpha_2 Tobin's\ Q + \alpha_2 ROA + \alpha_3 Stock\ return + \alpha_4 ROA\ volatility + \alpha_5 Stock\ return\ volatility + \alpha_6 Sales + \sum Year + \sum Industry + \varepsilon$$

静态面板实证方法

内生性与工具变量

Two-Stage-Least Squares (2SLS) Estimation

In this sub-section, we examine whether our results are robust to correcting for the possible endogeneity of stock liquidity by estimating a two-stage least squares regression.[8] Following Fang et al. (2009), we use two instruments that are correlated with stock liquidity, but are unlikely to be correlated with the error term. In particular, we use the lagged value of liquidity (*LagLiq*) and stock liquidity of the median firm in the industry (*IndLiq*) to capture the industry-driven component of a firm's liquidity. The use of lagged liquidity as an exogenous variable helps mitigate concerns that an unobservable in fiscal year *t* is correlated with both stock liquidity and compensation at time *t*. With regard to industry stock liquidity, Fang et al. point out that the portion of firm *i*'s liquidity that is correlated with the liquidity of its industry

静态面板实证方法

内生性与工具变量

is less likely to be correlated with unobservables that affect the outcome variable, which is firm *i*'s compensation in our case. The reduced-form equation for *Liq* (the linear combination of the exogenous variables in the system) becomes:

$$\begin{aligned} Liq = {} & \alpha_0 + \alpha_1 LagLiq + \alpha_2 IndLiq + \alpha_3 Tobin's\ Q + \alpha_4 ROA + \alpha_5 CFO \\ & + \alpha_6 Stock\ return + \alpha_7 ROA\ volatility + \alpha_8 CFO\ volatility \\ & + \alpha_9 Stock\ return\ volatility + \alpha_{10} LogSales + \sum Year + \sum Industry + \varepsilon \end{aligned} \quad (3)$$

静态面板实证方法

内生性与工具变量

Table 3 presents results of this instrumental variables approach. The coefficients on *LagLiq* and *IndLiq* in model (1) are positive and highly significant, suggesting that the instruments are highly correlated with stock liquidity. The next six models present the second-stage of the 2SLS regression where we regress *Cashcomp* and *LnDelta* on the predicted component of liquidity from the first stage. Consistent with our primary results from Table 2, the coefficient on *PrLiq* is negative and significant in all *Cashcomp* regressions and positive and significant in all *LnDelta* regressions. Thus, the relation between features of executive compensation and stock liquidity is robust to controlling for endogeneity using a two-stage least squares specification.

静态面板实证方法

内生性与工具变量

TABLE 2

Relation between Executive Compensation and Stock Liquidity

	Cashcomp			*LnDelta*		
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	0.680***	0.725***	0.954***	3.114***	3.082***	3.351*
	(0.049)	(0.078)	(0.164)	(0.301)	(0.839)	(1.844)
Liq	−0.091***	−0.069***	−0.041***	0.385***	0.303***	0.208***
	(0.003)	(0.004)	(0.007)	(0.019)	(0.035)	(0.033)
Tobin's Q	−0.035***	−0.023***	−0.018***	0.369***	0.363***	0.220***
	(0.002)	(0.003)	(0.004)	(0.012)	(0.018)	(0.013)
ROA	0.204***	0.195***	0.051	1.408***	1.616***	1.180***
	(0.029)	(0.035)	(0.042)	(0.178)	(0.205)	(0.143)
CFO	0.030	0.020	−0.035	−0.086	−0.197	0.178
	(0.028)	(0.032)	(0.037)	(0.171)	(0.206)	(0.134)
Stock return	0.030***	0.012**	0.018***	0.329***	0.368***	0.322***
	(0.005)	(0.005)	(0.005)	(0.028)	(0.025)	(0.019)
ROA volatility	−0.107**	−0.099*	−0.037	−1.163***	−1.267***	−0.747**
	(0.045)	(0.056)	(0.080)	(0.275)	(0.374)	(0.329)
CFO volatility	0.064	0.121*	−0.056	0.723**	0.724	0.864**
	(0.052)	(0.067)	(0.089)	(0.318)	(0.477)	(0.350)
Stock return volatility	−0.019***	−0.012*	−0.027**	0.173***	0.160***	0.119***
	(0.005)	(0.007)	(0.011)	(0.032)	(0.043)	(0.040)
LogSales	−0.057***	−0.051***	−0.019**	0.317***	0.348***	0.363***
	(0.002)	(0.002)	(0.008)	(0.009)	(0.017)	(0.039)
Year effects	Y	Y	Y	Y	Y	Y
Industry effects	Y	Y	Y	Y	Y	Y
Executive effects	N	N	Y	N	N	Y
Obs.	21,750	21,750	21,750	21,750	21,750	21,750
Adj. R^2		0.221	0.459		0.336	0.839

TABLE 3

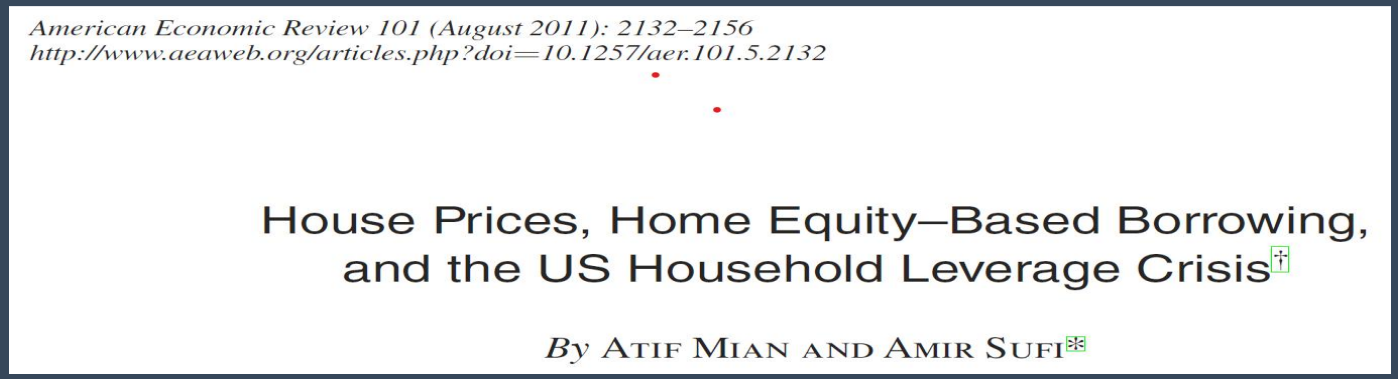
Two-Stage Least Squares (2SLS) Regressions

	1st stage	*2nd stage*					
	Liq	*Cashcomp*			*LnDelta*		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Const.	-.090	.584***	.651***	-.300***	3.922***	3.768***	8.392***
	(.065)	(.050)	(.080)	(.105)	(.288)	(.855)	(.650)
LagLiq	.741***						
	(.005)						
IndLiq	.270***						
	(.007)						
PrLiq		-.103***	-.079***	-.043***	.484***	.395***	.276***
		(.003)	(.005)	(.008)	(.020)	(.041)	(.036)
Tobin's Q	.019***	-.032***	-.022***	-.018***	.356***	.355***	.221***
	(.003)	(.002)	(.003)	(.004)	(.011)	(.018)	(.013)
ROA	.329***	.222***	.196***	.045	1.479***	1.601***	1.185***
	(.041)	(.029)	(.035)	(.041)	(.164)	(.205)	(.144)
CFO	-.089**	.028	.021	-.036	-.092	-.207	.185
	(.038)	(.027)	(.032)	(.037)	(.157)	(.206)	(.134)
Stock return	.078***	.030***	.011**	.019***	.319***	.372***	.320***
	(.007)	(.004)	(.005)	(.005)	(.026)	(.025)	(.018)
ROA volatility	.146**	-.083*	-.082	-.028	-1.375***	-1.428***	-.824**
	(.063)	(.044)	(.056)	(.080)	(.254)	(.375)	(.327)
CFO volatility	.040	.104**	.145**	-.050	.503*	.517	.763**
	(.069)	(.051)	(.068)	(.089)	(.294)	(.479)	(.350)
Stock return volatility	.046***	-.014***	-.009	-.027**	.151***	.131***	.106***
	(.008)	(.005)	(.007)	(.011)	(.030)	(.043)	(.039)
Log sales	.012***	-.055***	-.051***	-.018**	.309***	.340***	.347***
	(.002)	(.002)	(.002)	(.008)	(.009)	(.017)	(.039)
Year effects	Y	Y	Y	Y	Y	Y	Y
Industry effects	Y	Y	Y	Y	Y	Y	Y
Executive effects	N	N	N	Y	N	N	Y
Obs.	21,745	21,745	21,745	21,745	21,745	21,745	21,745
Adj. R²	.866		.221	.458		.339	.839

基准回归结果OLS与2SLS

静态面板实证方法

内生性与工具变量



- 房价上升，导致已有住房者借款行为变化，借款更多，但借款并不是为了购房新房、不是购买投资性住房、也不是为了偿还信用卡，而主要为了消费行为，从而导致家庭债务危机

静态面板实证方法

内生性与工具变量

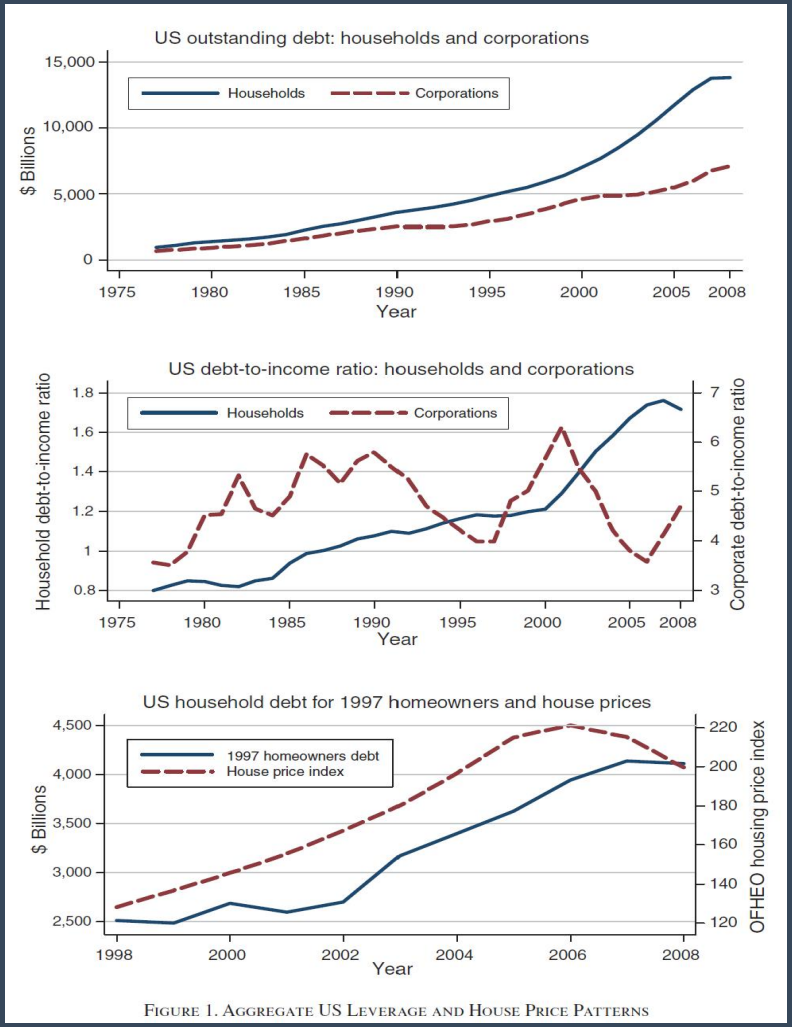


FIGURE 1. AGGREGATE US LEVERAGE AND HOUSE PRICE PATTERNS

- 从1978-2008，家庭债务增加，远多于公司债务
- 家庭债务收入比逐步上升，而公司债务收入比没有明显上升趋势
- 1997年拥有住房者债务与房价指数，两者有明显的趋势

静态面板实证方法

内生性与工具变量

Figure 2 and the estimate in column 1 of Table 2 motivate the following first-differenced instrumental variables (IV) specification:

$$(1)\quad LeverageGrowth0206_{izm} = \theta X_{izm} + \beta \widehat{HousePriceGrowth0206}_{zm} + \upsilon_{izm},$$

$$(2)\quad HousePriceGrowth0206_{zm} = \delta X_{izm} + \rho Elasticity_{m,1997} + \varepsilon_{izm},$$

- （2）为2SLS第一阶段，（1）为第二阶段。住房供给弹性（住房供给对住房需求的反应程度）作为工具变量，影响房价。
- 第二阶段，房价（拟合值）影响家庭债务增长率

静态面板实证方法

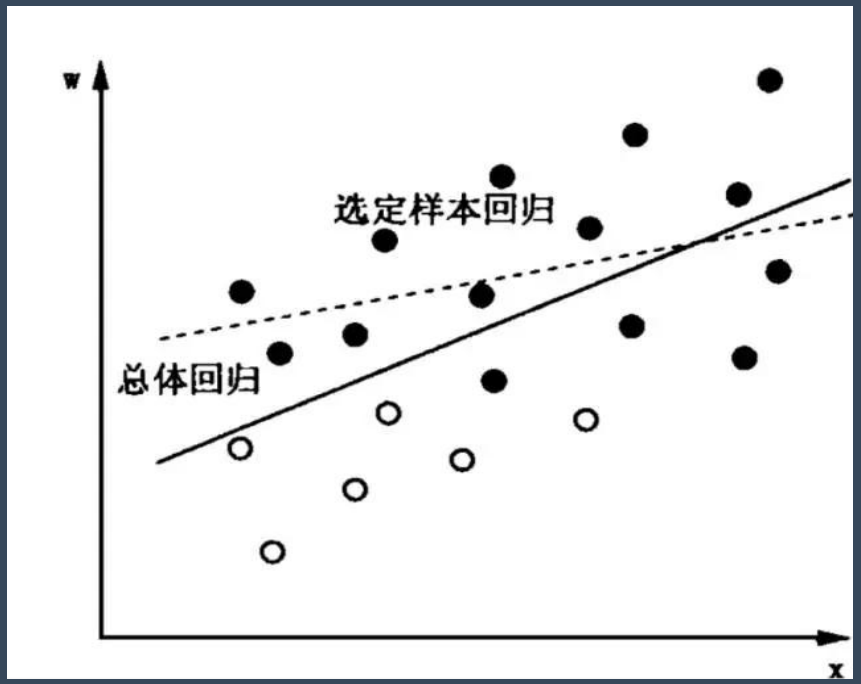
方法2 ： 两阶段选择模型

- 样本选择偏误所致的内生性问题

  - 选择偏误：财务危机预警以已通过审核申请者样本建立模型，忽略未通过审核申请者样本。然而以这些样本所建构出来的模型，无论是在变量的选取、权重的设计、对未来用于审核所有贷款申请人是否给予贷款的决策上，均容易产生偏误。未考虑样本选择偏误（sample selection bias）问题，故会影响模型的配适度与预测能力。

静态面板实证方法

方法2 ： 两阶段选择模型



- 白点加黑点是全样本的散点图，但由于样本选择问题，只选择了黑色散点所代表的样本。
- 实线是根据总样本（黑点加白点所代表的样本）回归所得，而虚线是根据选择的样本回归所得。两线斜率、截距明显不同，即样本选择会带来偏误。

静态面板实证方法

方法2 ： 两阶段选择模型

- 选择偏误的三种情形
- 情形1：只选择部分样本，无控制组 （扩大样本）
- 情形2：因变量存在缺失值即无法选择（用Heckman两阶段选择模型：heckman）；
- 情形3：因变量可能影响选择性的内生变量或自选择问题（用两阶段处理效应模型:treatreg）

静态面板实证方法

方法2 ： 两阶段选择模型

- 情形2：2001年07月09日中国台湾地区公布金融控股公司法，允许金融控股公司的成立；因变量为公司绩效=是否加入金控。传统思维：设定一个虚拟变量，加入=1，不加入=0，但这样产生选择偏误。
- 问题1：那些加入金控的银行本来就是比较好的，而那些没加入的本来财务状况就比较差——是加入金控提高业绩还是样本选择的差异导致就很难分清
- 问题2：可以知道那些已经加入金控的公司的业绩是好是坏，但没有办法知道这些公司如果没有加入的后果（换句话说，无法选择或因变量有部分样本是无法观测的）。 （思考问题的方式非常重要）

静态面板实证方法

方法2 ： 两阶段选择模型

- 情形3：因变量为投资（僵尸企业+非僵尸企业）=是否为僵尸企业（僵尸企业为1）+现金流+托宾Q。 然而，选择什么都不做的企业（减少投资或不投资），越有可能成为僵尸企业
- 因变量为公司投融资（或公司绩效）=官员是否视察+控制变量。但是，官员选择是否视察是内生的，公司绩效越高的公司，官员越有可能选择去视察。

静态面板实证方法

方法2 ： 两阶段选择模型

假设**被接受**母体分配不同于**被拒绝**母体分配，且Y_{1i}和Y_{2i}为随机变量：

$$Y_{1i} = \beta_1 X_{1i} + \varepsilon_{1i}$$

$$Y_{2i} = \beta_2 X_{2i} + \varepsilon_{2i}$$

- 选择性矫正因子(inverse Mill's ratio或Mill's lambda）

$$\lambda_i = \frac{\phi\left(\hat{\beta}_1 X_{1i}\right)}{\Phi\left(\hat{\beta}_1 X_{1i}\right)}$$

- $\hat{\beta}_1$ 为第一阶段回归系数，分子为正态分布概率密度函数，分母为累积概率密度函数

静态面板实证方法

方法2 ： 两阶段选择模型

第一阶段，先利用最大概似法估计二元Probit 模型，以获得β_1的估计值，并且对选择的样本中每一个估计值，计算其对应之**选择性矫正因子**，以矫正样本选择性偏误，并可观察两阶段间是否显著相互影响。

第二阶段，则将λ_i视为一项额外的解释变量，与解释变量X_{2i}一起建构第二阶段的Probit 模型，以得到β_2估算系数。若系数λ_i显著异于0，代表第二阶段会受到第一阶段影响，故λ_i可解释方程式中的样本选择偏误问题，即方程估计需要考虑样本选择偏差问题。

静态面板实证方法

方法2 ： 两阶段选择模型

- 软件实现
- 要解决的问题：reg var5 var7 var6 var4
- 第一阶段：prob var7 var6 var4 var3 var1 var2　　（var7为0-1变量）

  predict gw, xb

  g lambda=normalden(gw)/normal(gw) if var7==1

  replace lambda=-normalden(gw)/normal(-gw) if var7==0

  第二阶段：reg var5 var7 var6 var4 **lambda**
- 等同于treatreg var5 var6 var4, tr(var7= var6 var4 var3 var1 var2) twostep

静态面板实证方法

方法2 ： 两阶段选择模型



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Ownership concentration, foreign shareholding, audit quality, and stock price synchronicity: Evidence from China ☆

Ferdinand A. Gul [a], Jeong-Bon Kim [b,*], Annie A. Qiu [c]

$$SYNCH_{i,t} = \lambda_0 + \lambda_1 BIG4_{i,t} + \lambda_2 LOCAL_{i,t} + \sum_k \lambda_k CONTROL^k_{i,t} + (YearDummies) + (IndustryDummies) + \varepsilon_{i,t}, \quad (4)$$

静态面板实证方法

方法2 ： 两阶段选择模型

accounted for. This finding is consistent with the notion that ownership concentration beyond about 50% leads to the managerial entrenchment effect being dominated by the incentive alignment effect in China.

6. Robustness checks

6.1. Endogeneity

Thus far, we have not considered any potential self-selection bias that might arise from the fact that firms self-select their auditors. One cannot rule out the possibility, however, that low-synchronicity firms are more likely to choose Big 4 auditors. In such a case, examining the effect of Big 4 versus non-Big 4 auditor choice in a single-equation context may introduce a self-selection bias into our estimate of the Big 4 coefficient. We address this issue using a two-stage regression approach.

In the first stage, we estimate a probit regression in which the likelihood of Big 4 auditor choice, denoted by $Pr(BIG4)$, is regressed on a set of firm-specific variables that might influence the demand for high-quality auditors:

$$Pr(BIG4)_{i,t} = \delta_0 + \delta_1 LEV_{i,t} + \delta_2 M/B_{i,t} + \delta_3 SALES_{i,t} + \delta_4 SHRINC_{i,t} + \delta_5 LOSS_{i,t} + (YearDummies) + (IndustryDummies) + \varepsilon, \quad (5)$$

where, for firm i and year t, $Pr(Big4)$ is ex post coded as one for *Big* 4 clients, and zero otherwise; *SALES* denotes the dollar amount of sales; *SHRINC* equals one if the number of shares outstanding increases more than 10% in each sample year, and zero otherwise; *LOSS* equals one for loss firms, and zero otherwise; and *LEV*, *M/B*, and ε are as defined earlier.

- 经验结果是， 选择4大审计机构的公司，其股价同步性更低。然而，如果低股价同步性的公司更有可能选择4大审计机构，则产生选择偏误

静态面板实证方法

方法2 ： 两阶段选择模型

Table 6
Results of two-stage regressions to examine self-selection bias associated with auditor choice.

All variables except *SALES*, *SHRINC*, and *LOSS* are as defined in the Appendix. *SALES* represents the dollar amount of sales. *SHRINC* is an indicator variable that equal one if the number of shares outstanding increases more than 10% in each sample year and zero otherwise. *LOSS* is an indicator variable that equals one for loss firms and zero otherwise. *SYNCH* (1) is estimated using Eq. (1). *SYNCH*(1*a*,*b*,*c*) is estimated using Eqs. (1a)–(1c) for firms with A-shares only, firms with A- and B-shares, and firms with A- and H-shares, respectively. Numbers in parentheses represent *t*-values that are adjusted using standard errors corrected for clustering at the firm level. The superscripts a, b, and c denote the 1%, 5%, and 10% levels of significance, respectively.

	Section A First-stage probit regression	Section B Heckman approach with inverse Mills ratio (*Lamda*) included				Section C Two-stage least square (2SLS) approach with the predicted likelihood of auditor choice			
	(1) Dependent variable =*Pr(BIG4)*	(2) Dependent variable =*SYNCH*(1)		(3) Dependent variable =*SYNCH*(1*a*,*b*,*c*)		(4) Dependent variable =*SYNCH*(1)		(5) Dependent variable =*SYNCH*(1*a*,*b*,*c*)	
Panel A									
TOPHOLD^2			−0.992		−1.045		−0.738		−0.765
			(−2.60)^a		(−2.71)^a		(−2.00)^b		(−2.04)^b
TOPHOLD			1.012		1.031		0.828		0.827
			(2.86)^a		(2.87)^a		(2.44)^b		(2.40)^b
TOPGOV			0.111		0.106		0.093		0.085
			(3.14)^a		(2.94)^a		(2.70)^a		(2.46)^b
HSHARE			−0.384		−0.409		−0.263		−0.277
			(−4.42)^a		(−4.64)^a		(−3.19)^a		(−3.33)^a
BSHARE			−0.216		−0.187		−0.173		−0.141
			(−4.70)^a		(−4.18)^a		(−3.82)^a		(−3.20)^a
BIG4		−0.113	−0.057	−0.144	−0.088				
		(−3.31)^a	(−1.72)^c	(−4.03)^a	(−2.55)^b				
PredBIG4						−1.595	−1.412	−1.774	−1.593
						(−8.83)^a	(−7.59)^a	(−9.91)^a	(−8.60)^a
Panel B									
LOCAL		0.111	0.083	0.123	0.093	0.051	0.035	0.059	0.042
		(3.38)^a	(2.49)^b	(3.52)^a	(2.65)^a	(1.56)	(1.07)	(1.71)^c	(1.22)
VOL		−0.093	−0.110	−0.094	−0.110	−0.112	−0.121	−0.114	−0.122
		(−8.98)^a	(−8.78)^a	(−9.12)^a	(−8.70)^a	(−10.15)^a	(−9.43)^a	(−10.36)^a	(−9.40)^a
SIZE		0.064	0.075	0.043	0.057	0.033	0.041	0.009	0.019
		(3.82)^a	(4.44)^a	(2.42)^b	(3.13)^a	(2.10)^b	(2.48)^b	(0.56)	(1.08)
LEV	0.007	0.002	0.002	0.001	0.001	0.007	0.007	0.007	0.006
	(8.05)^a	(0.67)	(0.88)	(0.26)	(0.39)	(1.85)^c	(1.96)^c	(1.43)	(1.47)
STDROA		−0.004	−0.008	−0.001	−0.005	0.001	−0.004	0.004	0.001
		(−0.29)	(−0.70)	(−0.08)	(−0.45)	(0.04)	(−0.29)	(0.29)	(0.01)
M/B	−0.004	−0.052	−0.059	−0.058	−0.065	−0.117	−0.115	−0.130	−0.127
	(−37.11)^a	(−4.96)^a	(−5.93)^a	(−5.53)^a	(−6.50)^a	(−9.65)^a	(−9.57)^a	(−10.71)^a	(−10.64)^a
INDNUM		0.011	−0.033	−0.054	−0.094	−0.003	−0.036	−0.069	−0.098
		(0.09)	(−0.27)	(−0.45)	(−0.78)	(−0.02)	(−0.30)	(−0.58)	(−0.81)
INDSIZE		0.025	0.026	0.024	0.025	0.031	0.031	0.031	0.031
		(0.59)	(0.60)	(0.61)	(0.63)	(0.73)	(0.73)	(0.80)	(0.79)
SALES	0.001								
	(4.53)^a								
SHRINC	−0.011								
	(−1.52)								
Lamda		−0.007	−0.008	−0.009	−0.009				
		(−1.33)	(−1.53)	(−1.55)	(−1.69)^c				
LOSS	−0.054								
	(−6.93)^a								
Constant	0.131	−2.492	−2.635	−1.850	−2.053	−1.684	−1.808	−0.963	−1.140
	(11.30)^a	(−2.55)^b	(−2.68)^a	(−2.00)^b	(−2.19)^b	(−1.73)^c	(−1.84)^c	(−1.05)	(−1.22)
Industry dummies	Included	Included	Included	Included	Included	Included	Included	Included	Included
Year dummies	Included	Included	Included	Included	Included	Included	Included	Included	Included
N	5,994	5,994	5,994	5,994	5,994	5,994	5,994	5,994	5,994
Adj. R^2 (pseudo R^2)	(23.72%)	34.87%	36.43%	36.09%	37.40%	36.98%	38.02%	38.43%	39.22%

静态面板实证方法

方法2 ： 两阶段选择模型



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Financial dependence and innovation: The case of public versus private firms☆

Viral Acharya[a], Zhaoxia Xu[b,*]

[a] Department of Finance, Stern School of Business, New York University, New York, NY, USA
[b] Department of Finance & Risk Engineering, Tandon School of Engineering, New York University, New York, NY, USA

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ABSTRACT

In this paper, we examine the relation between innovation and a firm's financial dependence using a sample of privately held and publicly traded US firms. We find that public firms in external finance dependent industries spend more on research and development and generate a better patent portfolio than their private counterparts. However, public firms in internal finance dependent industries do not have a better innovation profile than private firms. The results are robust to various empirical strategies that address selection bias. The findings indicate that the influence of public listing on innovation depends on the need for external capital.

□与非上市公司相比，外部融资依赖行业中的上市公司投入更多研发支出和专利产出；然而，与非上市公司相比，内部融资依赖行业中的上市公司创新行为并没有显著多。

机制：外部融资依赖行业的上市公司，即内部融资不足，上市后能缓解融资约束，有利于创新；而内部现金流充足的公司，代理成本高，降低创新效率，短视行为也会抑制创新。

静态面板实证方法

方法2 ： 两阶段选择模型

$$Y_{i,k,t} = \alpha + \beta Public_{i,k,t} + \gamma X_{i,k,t-1} + \eta_k + \zeta_t + \varepsilon_{i,k,t}, \quad (3)$$

- *Y*为公司创新活动，RD投入，专利；*Public* 为虚拟变量，上市公司为1，非上市公司为0；

- 然而，公司是否成为上市公司并不是随机的，需要控制同时影响公司创新和公司上市决定的不可观测的因素。

$$Public_i = \begin{cases} 1 & \text{if} \quad Public_i^* > 0 \\ 0 & \text{if} \quad Public_i^* \leq 0 \end{cases} \quad Public_i^* = \pi + \delta Z_i + \upsilon_i \quad (4)$$

- *Z*为影响公司决定是否上市的公司特征变量

$$\begin{aligned} Y_{i,k,t} = {} & \alpha + \beta Public_i + \delta EFD_{i,k} + \theta Public_i \times EFD_{i,k} \\ & + \gamma X_{i,k,t-1} + \lambda X_{i,k,t-1} \times EFD_{i,k} + \phi Mills_i + \varepsilon_{i,k,t}, \end{aligned} \quad (5)$$

静态面板实证方法

方法2 ： 两阶段选择模型

Panel A: Treatment effect model first step

Variable	All	EFD industries	IFD industries
Capex	0.7917***	0.9105***	0.9592
	[0.1971]	[0.2083]	[0.6897]
S.Growth	−0.0646**	−0.0882***	0.0904
	[0.0266]	[0.0286]	[0.0807]
ROA	−0.5451***	−0.6769***	0.1305
	[0.0858]	[0.0915]	[0.2625]
Sales	−0.0023	0.0059	−0.0681***
	[0.0071]	[0.0077]	[0.0188]
Leverage	−1.6542***	−1.6941***	−1.5700***
	[0.0430]	[0.0473]	[0.1057]
Intensity	0.1570***	0.3182***	0.0625
	[0.0510]	[0.0598]	[0.1029]
EFD	0.1679***		
	[0.0542]		
N	12,253	10,137	2,116

Variable	*ln(R&D)*	*ln(Patent)*	*ln(Citations)*	*Originality*	*Generality*
Panel B: External finance dependent industries					
Public	0.7535***	0.5072***	0.5109***	0.1301***	0.1709***
	[0.0699]	[0.0511]	[0.0589]	[0.0371]	[0.0502]
Mills	−0.3759***	−0.1745***	−0.1812***	−0.0315	−0.0623**
	[0.0421]	[0.0309]	[0.0356]	[0.0207]	[0.0280]
N	10,137	10,137	10,137	2,480	2,480
Panel C: Internal finance dependent industries					
Public	0.0538	-0.0844	-0.0911	-0.0495	-0.2141
	[0.0602]	[0.0644]	[0.0902]	[0.1212]	[0.1540]
Mills	-0.0180	0.0787**	0.0832	0.0487	0.0891
	[0.0370]	[0.0395]	[0.0554]	[0.0734]	[0.0930]
N	2,116	2,116	2,116	192	192
Panel D: Industry and size matched sample					
EFD × *Public*	0.4996***	0.5538***	0.5340***	0.2422***	0.1122
	[0.0845]	[0.0633]	[0.0746]	[0.0608]	[0.0906]
N	12,253	12,253	12,253	2,672	2,672
Panel E: Age, year, and size matched EFD and IFD pairs					
EFD × *Public*	0.2663**	0.3929***	0.4642***	0.2485**	0.1202
	[0.1437]	[0.0740]	[0.0976]	[0.1215]	[0.1830]
N	3,531	3,531	3,531	525	525
Panel F: Age, year, and R&D matched EFD and IFD pairs					
EFD × *Public*	0.9247***	0.4452***	0.4207***	0.2035	0.3482*
	[0.1094]	[0.0873]	[0.0914]	[0.2033]	[0.2104]
N	2,533	2,533	2,533	356	356
Panel G: Exclude M&As					
EFD × *Public*	0.1741*	0.3429***	0.4533***	0.3315***	0.2113**
	[0.0892]	[0.0640]	[0.0846]	[0.0701]	[0.1015]
N	8,703	8,703	8,703	1,715	1,715

静态面板实证方法

方法 3：倾向评分匹配

- Propensity Score Match (PSM)
- 考虑国有企业效率问题。因变量为国有企业投资效率=国有企业的规模+国有企业负债率+…。
  正确做法：处理组为国有企业，控制组为非国有企业（以及国外的国有企业和非国有企业），在同一纬度上进行比较才有意义。



- 考虑中国上市公司长期负债比例过低问题，是否为中国独有的现象。处理组为中国上市公司，控制组为外国上市公司，在同一纬度上进行比较有意义。

静态面板实证方法

方法 3：倾向评分匹配

- 两对被抱错双胞胎的不同人生轨迹



两对被抱错双胞胎的不同人生轨迹

- 这是一个关于两对双胞胎的故事。那是1988年，有两对同卵双胞胎在哥伦比亚波哥大的一个医院同时出生。然而，医院的一个失误，把他们4个搞混了，两个家庭以为是异卵双胞胎，就这样各领了一个自己的和一个别人的孩子回家抚养长大。两对兄弟的未来历程，也因此发生了翻天覆地的变化。20多年后，他们长成了这样。
- Bernal家的Jorge和Carlos，他们在城市里长大，于是，一个成了工程，一个成了会计。Caas家的William和Wilber，他们在农村长大，于是都成了屠夫，打理着家里的一家肉铺。
- 两对双胞胎就这么互相不知道彼此度过了20多年。直到有一天，Jorge的一个朋友到William的肉铺里买肉，突然看到了正在卖肉的William，Jorge的同事完全不敢相信，以为是Jorge在这里卖肉。然而William表示，他不是Jorge，他也不知道Jorge是谁。肉店里别人也说，William工作很努力，几乎除了睡觉之外从来不离开肉铺。



两对被抱错双胞胎的不同人生轨迹

- 如果没有抱错，右边的William就不会成为一个屠夫。他会在城里上学，接受教育，像Jorge或Carlos一样，成为工程师或者会计，而不是像现在这样在农村的农场里工作。
- William从小就很努力，想摆脱自己在乡下的困境。他决定去参军，因为表现突出，他甚至得到了上军官学校的奖学金。但是，就在即将被接受时，他发现，奖学金的先决条件是必须要有中学毕业证书，他没有，他12岁时就辍学了，因为离他最近的中学也在走路5小时之外的地方。如果他要去上学，家里就要给他出住宿费等，还少了个壮劳力。没办法，只能让他在家里工作。
- 就这样，两兄弟的道路越走越远。William和Wilber生活在穷困的乡下，而Jorge和Carlos虽然生活在城里，但是他们家境也不是特别好。两个家庭家境都不好，但在城里的两兄弟有机会接受了教育。于是，这四人的职业轨迹发生了如此大的变化。然而，他们只能选择接受这一切。





静态面板实证方法

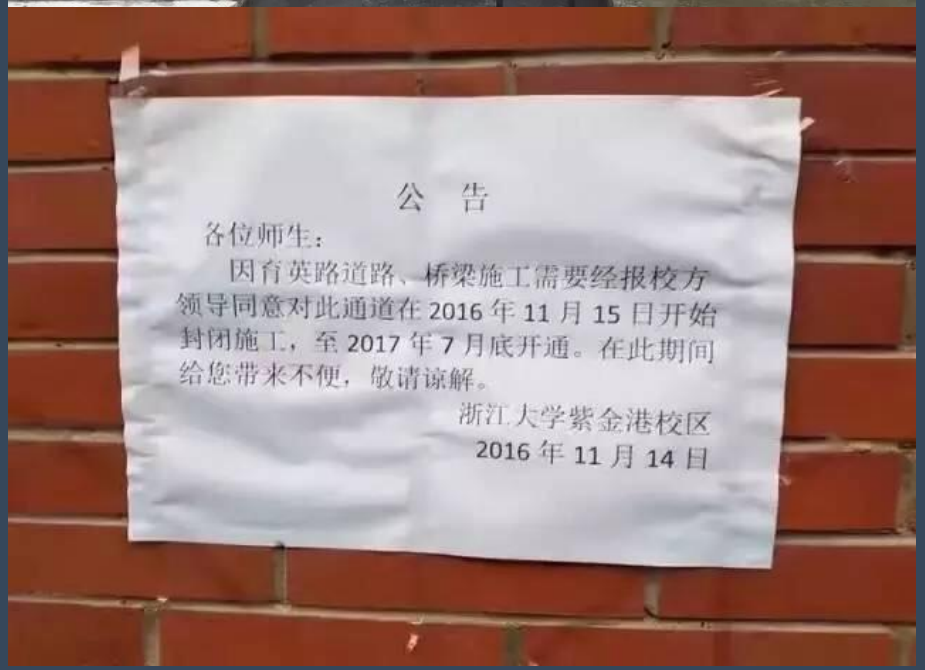
方法 3：倾向评分匹配

- **PSM的原理和作用**
- 设置处理组与控制组
- **控制组**中并不是所有的个体都适合拿来对比，而是应该寻找出那些跟处理组(treatment)组相近的样本，也即排除其他因素的影响；
- 要做到上一步，需要配对，PSM的思想是通过对比两组样本的其他维度信息，筛选出在其他维度相似的样本，使其具有可比性。


开门时间
6:00-23:00
严禁攀爬


汇鑫小吃
烤红薯 绿豆汤
烤玉米 3元/杯
西安凉
绿豆汤
3元/杯
绿豆汤
茶叶蛋
炒粉干
凉皮
凉面
美团外卖
吃货新时代!
手抓饼
原味 3元
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原味 2.5元
加培根 2元
加芝士 2元
加鸡柳 2元
加骨肉相连 2元
加素菜 每份1元
免费加生菜




公 告
各位师生：
因育英路道路、桥梁施工需要经报校方领导同意对此通道在2016年11月15日开始封闭施工，至2017年7月底开通。在此期间给您带来不便，敬请谅解。
浙江大学紫金港校区
2016年11月14日

静态面板实证方法

方法 3：倾向评分匹配

- 健康=锻炼+吃喝+睡觉+学习压力+堕落街
- 所有解释变量都可能存在内生性问题。例如，
- 外生事件，非内生：以道路整治改造封闭为外生事件，处理组为到堕落街吃喝的同学，控制组为不到堕落街吃喝的同学。
- 处理组为堕落街吃喝的同学，但政策出台后不再继续“堕落”的同学。控制组为堕落街吃喝的同学，但政策出台后通过各种渠道继续“堕落”的同学。

静态面板实证方法

方法 3：倾向评分匹配

- **PSM的一般顺序**
- 先选取初始的处理组(treatment=1)和控制组(treatment=0)，并以该虚拟变量作为因变量，选取可能影响它的自变量对其进行回归分析(logit或者probit 模型)：
- 进行logit或者probit估计后求出每个观测的倾向分值（pscore），即实验组的概率p，根据p重新分组；
- 检验匹配的效果，确保匹配后两组样本在其他维度没有显著差别；
- 用匹配后的新样本进行下一步的分析；

静态面板实证方法

方法 3：倾向评分匹配

- 软件实现
- 一步到位的命令：psmatch2 var7 varlist，n(1)
- var7是一个虚拟变量，指最初的处理组（1）和控制组（0）；varlist指影响var7的变量；n(1)指匹配的选项，这里指1对1匹配，也可以选择其他形式
- 运行 psmatch2命令后，软件会出现以下信息（相应含义见下一张PPT），其中主要关注pscore和weight
- 根据weight筛选出新的样本

_pscore	_treated	_support	_weight	_id	_n1	_nn	_pdif
.18035039	Untreated	On support	1	4779	.	0	.
.18111245	Untreated	On support	.	7973	.	0	.
.1812049	Untreated	On support	.	8675	.	0	.
.18105737	Untreated	On support	.	7595	.	0	.
.18122554	Untreated	On support	1	8878	.	0	.
.18120162	Untreated	On support	.	8647	.	0	.

静态面板实证方法

方法 3：倾向评分匹配

- treated 表示处理组（不等于空值，var7为1）,untreated控制组；
- support：是否进行了匹配；
- pscore：倾向匹配得分；
- weight: 匹配频率，有些控制组样本会被多次配比；
- id：配对过程系统给每个观测分配新的ID;
- n1：第几次配对的控制组的ID；
- nn：每个处理组所配对到的控制组数量；
- pdif：处理组合控制组之间probit回归的概率值之差；

静态面板实证方法

方法 3：倾向评分匹配

- 软件实现
- 第1步：用psmatch2命令进行匹配

psmatch2 var7 var4 var5,n(1)

n(1)表示实验组和控制组是一一配对

- 第2步: 生成配对后新的样本,命名新的分组变量var7_new

gen var7_new=0

replace var7_new=1 if var7==1 & _weight!=.

replace var7_new=. if _weight==.

!=不等于， weight表示匹配的权重，空值表示没有被匹配到

静态面板实证方法

方法 3：倾向评分匹配

- 第3步：匹配效果

  probit var7_new var4 var5

  对匹配后的样本进行回归，要求控制变量没有显著影响

- 附加：匹配前后实验组和控制组的数量变化，一般会有所减少

  count if var7==1

  count if var7==0

  count if var7_new==1

  count if var7_new==0

- **psmatch2 var7 var4 var5,n(1) 一对一匹配**
- **psmatch2 var7 var4 var5,**kernel ate ties common
- **psmatch2 var7 var4 var5,**Outcome(var4) kernel ate ties common

(kernel 优于一对一匹配？)

静态面板实证方法

方法3：倾向评分匹配

- 结果检验
- 一般配对后要求两组公司的控制变量没有显著差别，对匹配后的样本进行probit模型回归，要求控制变量对新的treatment没有显著影响，如Fang et al.(JF，2014),

Panel B: Prematch Propensity Score Regression and Postmatch Diagnostic Regression

	Dummy = 1 if in treatment group; 0 if in control group	
	(1)	(2)
Dependent Variable	Prematch	Postmatch
$ILLIQ_{-1}$	0.848***	0.043
	(0.072)	(0.100)
LN_MV_{-1}	0.178***	0.006
	(0.033)	(0.046)
$RDTA_{-1}$	−0.435	0.315
	(0.425)	(0.582)
ROA_{-1}	0.774***	−0.041
	(0.178)	(0.245)
$PPETA_{-1}$	0.871***	0.052
	(0.193)	(0.287)
LEV_{-1}	0.051	−0.024
	(0.154)	(0.217)
$CAPEXTA_{-1}$	−2.135***	−0.255
	(0.562)	(0.789)
$HINDEX_{-1}$	−0.118	−0.083
	(0.388)	(0.530)

Match后变量的系数不再显著

静态面板实证方法

方法3：倾向评分匹配

Do U.S. Firms Hold More Cash than Foreign Firms Do?

Lee Pinkowitz
Georgetown University

René M. Stulz
The Ohio State University

Rohan Williamson
Georgetown University

From 1998 to 2011, U.S. firms held more cash on average (but not at the median) than similar foreign firms (foreign twins) did. The average difference in cash holdings does not increase after 2008, and it is driven by highly R&D-intensive U.S. firms. Because there are almost no similarly R&D-intensive foreign firms, mean comparisons involving these U.S. firms are not reliable. Without these U.S. firms, neither U.S. multinational firms nor purely domestic U.S. firms hold more cash than their foreign twins do. Country characteristics have negligible explanatory power for differences in cash holdings between U.S. firms and their foreign twins. (*JEL* G30, G31, G32)

Received April 17, 2014; accepted August 4, 2015 by Editor David Denis.

□大多数研究已经表明，美国公司持有更多的现金，然而这种现象是美国独有的现象，还是普遍存在的现象？。要回答的是：与同样的外国公司相比，美国公司是否持有更多现金？

静态面板实证方法

方法3：倾向评分匹配

To compare the cash holdings of U.S. firms with the cash holdings of similar foreign firms, we proceed as follows. We want the firms to be similar in firm characteristics that are associated with cash holdings in the literature. We use the propensity-score matching technique. This technique has been used before in comparing U.S. firms to foreign firms (see, for instance, Aggarwal et al. 2009; Bartram, Brown, and Stulz 2012). The propensity score of a firm is equal to the probability that the firm with given characteristics belongs to a particular group, called the treatment. The treatment in this case is whether the firm is a U.S. firm. For each propensity-score match, we use the determinants of cash holdings based on BKS as the covariates in the propensity-score regression along with higher-order terms of the covariates if doing so improves the quality of the matching.[6] Each year, for each U.S. firm, we select as its twin the foreign firm that has the closest propensity score without replacement. We require common

- 处理组：美国公司；
- 控制组：国外公司；
- 第一阶段probit回归的因变量为是否为美国公司，解释变量同BKS（JF,2009）

静态面板实证方法

方法 3：倾向评分匹配

Table 4
Comparison of the annual cash holdings of U.S. Firms and their foreign twins

Year	Median difference (U.S. minus foreign)	Mean difference (U.S. minus foreign)	Possible matches	Matched pairs	% Matched
1998	−0.012	0.002	3,365	1,348	40%
1999	−0.010	0.001	3,256	1,388	43%
2000	−0.014*	−0.003	3,095	1,595	52%
2001	−0.017**	−0.007	2,883	1,533	53%
2002	−0.009	0.006	2,781	1,503	54%
2003	−0.012	0.006	2,816	1,607	57%
2004	−0.017	0.001	2,831	1,671	59%
2005	−0.005	0.015*	2,794	1,749	63%
2006	−0.005	0.020**	2,770	1,904	69%
2007	−0.006	0.019**	2,742	2,018	74%
2008	−0.010	0.010	2,617	1,953	75%
2009	−0.002	0.014	2,614	2,123	81%
2010	−0.002	0.013	2,569	2,113	82%
2011	−0.006	0.010	2,504	2,061	82%
Fama-Macbeth	−0.009***	0.008**	39,637	24,566	62%

- each year the typical U.S. firm holds less cash than its foreign twin does.
- Only two median differences are significant, in 2000 and 2001. There is a trace of an increase in the median difference over time, but this increase is insignificant.
- number of U.S. firms to be matched falls over time as the number of listed firms decreases in the US.

静态面板实证方法

方法 3：倾向评分匹配

$$\left(\frac{Cash}{Assets}\right)_{i,c} = \alpha + \delta US_{i,c} + \beta_1 CF_{i,c} + \beta_2 Ind\,Vol_{i,c} + \beta_3 MB_{i,c} + \beta_4 Size_{i,c} + \beta_5 NWC_{i,c}$$
$$+ \beta_6 Capex_{i,c} + \beta_7 Leverage_{i,c} + \beta_8 RD_{i,c} + \beta_9 Dividend_{i,c}$$
$$+ \beta_{10} Acquisitions_{i,c} + \beta_{11} Debt\,Issues_{i,c}$$
$$+ \beta_{12} Equity\,Issues_{i,c} + \varepsilon_{i,c} \quad \forall\ year\ t \qquad (1)$$

- 每年进行一次回归

静态面板实证方法

方法 3：倾向评分匹配

不同国家公司数量差别大，用加权OLS

cash/assets ratio has a skewed distribution 中位数回归

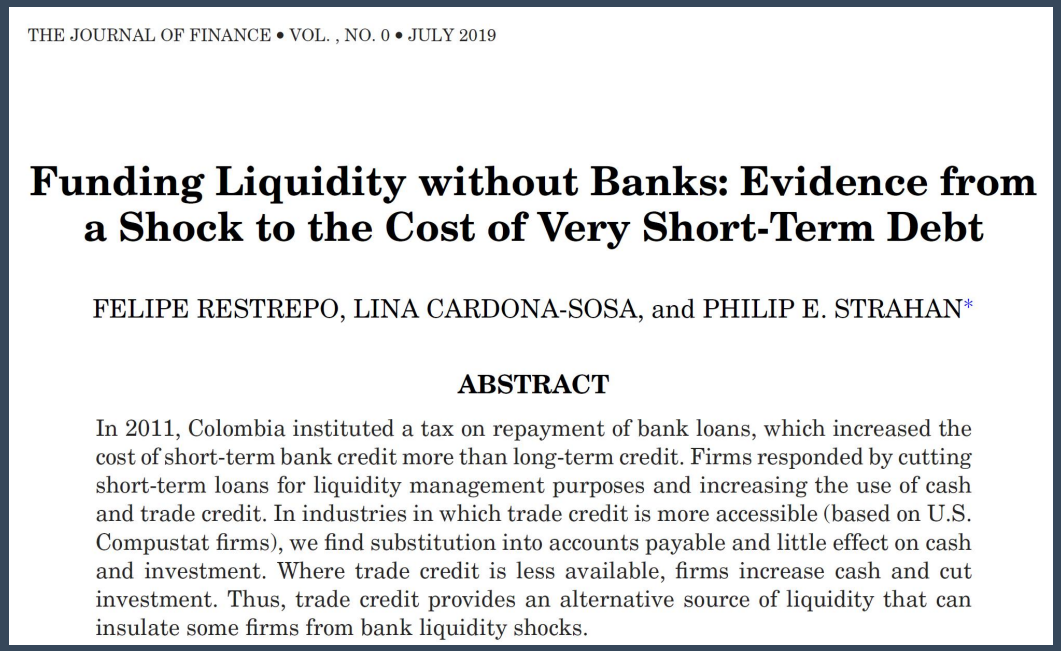
Table 5
Cash holdings in the United States and with MNC firms: Annual cross-regressions on the full unmatched sample

	(1) OLS U.S. vs. foreign	(2) WLS U.S. vs. foreign	(3) Quantile U.S. vs. foreign
1998	0.014	0.030***	−0.007
1999	0.011	0.023**	−0.004
2000	0.007	0.013	−0.009
2001	0.000	−0.010	−0.014*
2002	0.012	0.005	−0.002
2003	0.008	−0.004	−0.005
2004	0.010	−0.002	−0.004
2005	0.017**	0.007	0.002
2006	0.022***	0.013	0.007
2007	0.024***	0.009	0.004
2008	0.018**	0.009	−0.001
2009	0.026***	0.017	0.012
2010	0.021**	0.022**	0.010
2011	0.018**	0.024**	0.003
Fama-Macbeth	0.015***	0.011**	−0.001

- positive and significant in the last 7 years of the sample, and insignificant in the first 7 years.

静态面板实证方法

方法 3：倾向评分匹配



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Funding Liquidity without Banks: Evidence from a Shock to the Cost of Very Short-Term Debt

FELIPE RESTREPO, LINA CARDONA-SOSA, and PHILIP E. STRAHAN*

ABSTRACT

In 2011, Colombia instituted a tax on repayment of bank loans, which increased the cost of short-term bank credit more than long-term credit. Firms responded by cutting short-term loans for liquidity management purposes and increasing the use of cash and trade credit. In industries in which trade credit is more accessible (based on U.S. Compustat firms), we find substitution into accounts payable and little effect on cash and investment. Where trade credit is less available, firms increase cash and cut investment. Thus, trade credit provides an alternative source of liquidity that can insulate some firms from bank liquidity shocks.

- 2011年，哥伦比亚对银行短期贷款的还款征税，使短期银行信贷的成本比长期信贷增加了更多。作为回应，公司削减了用于流动性管理目的的短期贷款，并增加了现金和商业信用的使用。 在更容易获得商业信用的行业中，公司应付账款增加，而且对现金和投资的影响很小。而在缺乏商业信用的行业，企业增加现金并削减投资。 因此，商业信用提供了另一种流动性来源，可以使一些公司免受银行流动性冲击的影响。

静态面板实证方法

方法 3：倾向评分匹配

- 图5可以看出，2007-2010年，大约40%的大公司的短期债务几乎为零，而最高的五分之一的公司短期债务/总资产非常高。

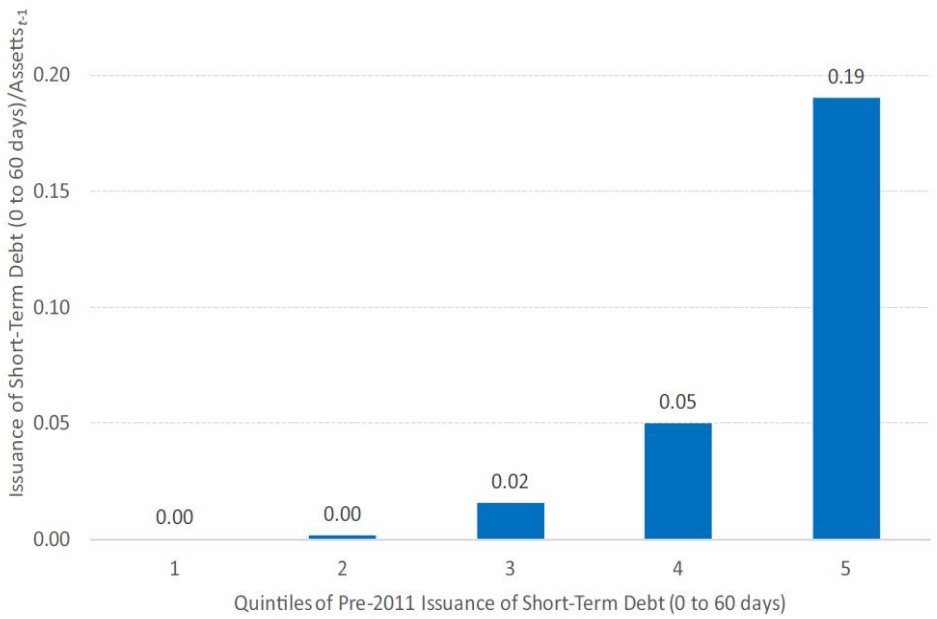


Figure 5. Quintiles of issuance of short-term debt (0 to 60 days) in the Pre-2011 period. This figure presents the average issuance of short-term debt (0 to 60 days) scaled by beginning-of-year assets for the five quintiles constructed based on the total issuance of short-term debt over total beginning-of-year assets in the preshock period (2007 to 2010). (Color figure can be viewed at wileyonlinelibrary.com)

静态面板实证方法

方法3：倾向评分匹配

- 因此，将最高的五分之一的公司定义为**处理组**。 **控制组**为在图5中最低的两个五分位数中的公司，因为它们的销售规模大致相当，但银行短期债务几乎为零。

- 使用**近邻匹配**。处理组和控制组公司的匹配是在冲击发生的前一年（2010年）进行的，对每个处理公司使用三个控制公司匹配。要求企业在冲击前后至少要有一个观察结果。

Panel A. Probit Estimation

	Dependent Variable: Dummy High Access to Short-Term Debt (0 to 60 days) Pre-2011
	(1)
Constant	-6.175***
	(0.584)
$Ln(Assets)_{t-1}$	0.638***
	(0.060)
$Ln(BV\ Equity)_t$	-0.396***
	(0.051)
$Asset\ Tangibility_t$	-0.323
	(0.220)
$Ln(Firm\ Age)_t$	0.225***
	(0.046)
$Asset\ Growth_t$	0.249*
	(0.130)
$Asset\ Turnover_t$	0.190***
	(0.023)
Observations	1,876
Pseudo R^2	0.080

Panel B. Pairwise Comparisons of Covariates used in Matching Procedure

	Unmatched Sample (Pre-2011 Period Means)			Matched Sample (Pre-2011 Period Means)		
	Control (Pre-Match)	Treatment	*t*-test Diff. in means (p-values)	Control (Post-Match)	Treatment	*t*-test Diff. in means (p-values)
Ln(Assets)	17.0	17.1	0.017	17.1	17.1	0.975
Ln(BV Equity)	16.1	16.0	0.142	16.1	16.0	0.738
Asset Tangibility	17.0%	14.3%	0.000	14.6%	14.3%	0.743
Firm Age	22.9	24.1	0.151	25.6	24.1	0.167
Asset Growth	13.7%	14.2%	0.472	14.5%	14.3%	0.830
Asset Turnover	1.9	2.5	0.000	2.4	2.6	0.158

静态面板实证方法

方法 4： 双重差分

- 双重差分模型(difference-in-difference，DID) 比较某一政策对不同组别的样本的影响，检验某一外生事件产生的影响。例如：金融危机、政策变化等
- **模型形式1（不含固定效应）：**

$$Y_{i,t}=\beta_0+\beta_1 Treat_i+\beta_2 After_t+\beta_3 Treat_i*After_t+e_{i,t}$$

Treat: 处理组（受到事件影响的样本）=1，控制组（不受事件影响的样本）=0

After: 事件发生后=1，否则=0

- **模型形式2（含固定效应）：**

$$Y_{i,t}=\beta_0+\beta_3 Treat_i*After_t+m_t+m_i+e_{i,t}$$

静态面板实证方法

方法 4：双重差分

	事件发生前	事件发生后	Difference
处理组	Y_{t1}	Y_{t2}	$\Delta Y_t = Y_{t2} - Y_{t1}$
控制组	Y_{c1}	Y_{c2}	$\Delta Y_c = Y_{c2} - Y_{c1}$
Difference			$\Delta\Delta Y = \Delta Y_t - \Delta Y_c$

静态面板实证方法

方法 4： 双重差分

- $Y_{i,t}=\beta_{0+}\beta_1\ Treat_i+\ \beta_2 After_t+\ \beta_3\ Treat_i*After_t+e_{i,t}$

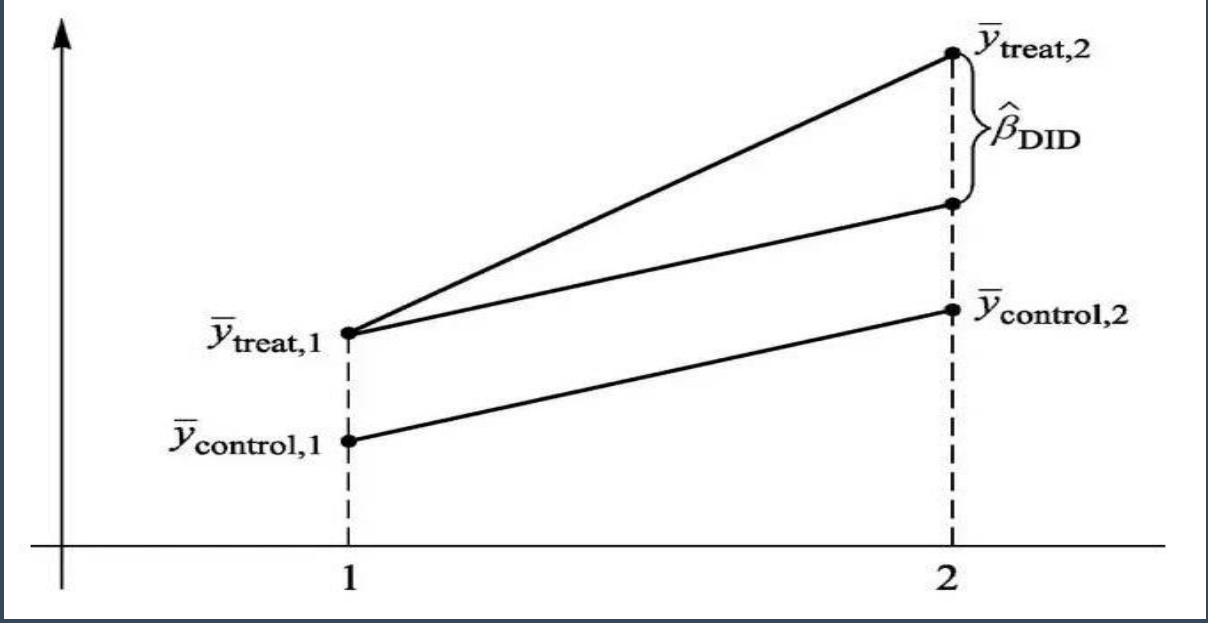
	事件发生前	事件发生后	Difference
处理组	β0+ β1	β0+ β1+ β2+ β3	ΔYt = β2+ β3
控制组	β0	β0+ β2	ΔYc= β2
Difference			ΔΔY = β3

- 模型当中关键看β_3是否显著

静态面板实证方法

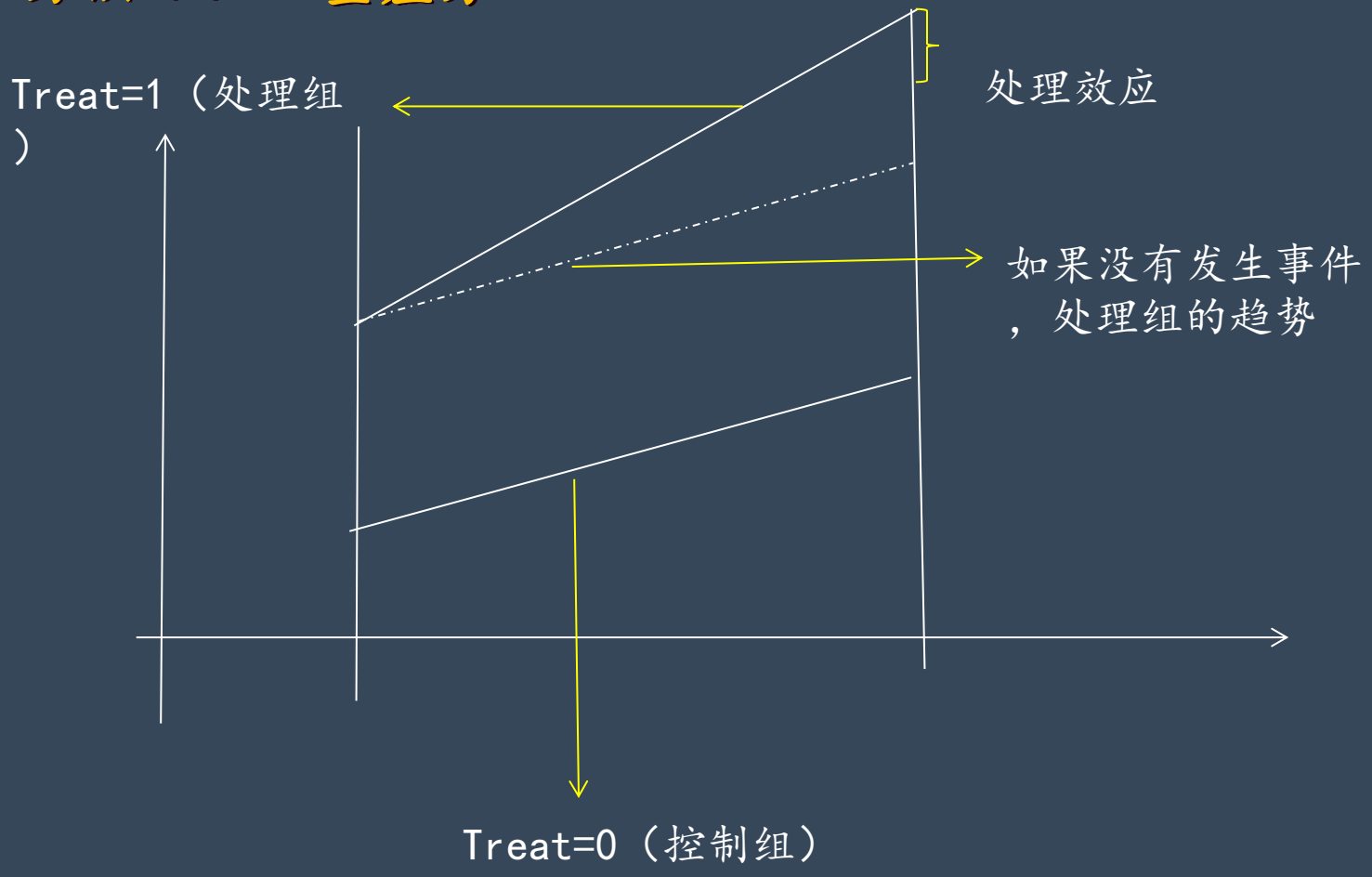
方法 4：双重差分

- DID的前提：平行趋势检验
- 处理组如果未受到政策干预，其时间效应或趋势应与控制组一样，即“平行趋势”（parallel trend）或“共同趋势”（common trend）。也即假定如果没有发生事件A，处理组与控制组的因变量（Y）应该有相同的趋势，即相同的增长率



静态面板实证方法

方法4：双重差分



静态面板实证方法

方法 4： 双重差分

DID 模型的软件实现

- 通过PSM匹配样本后，接下来进行DID模型的回归。先生成时间变量：after, 事件发生（2007年）后=1， 发生之前=0
- gen after=0
- replace after=1 if var2>2007
- step1: 通过描述性统计先观察不同组别的公司在冲击前后利润是否有显著不同
- diff var6, t(var7) p(after)
- step2: 回归分析
- reg var6 var7 after c.var7#c.after var4 var5
- reg var6 c.var7#c.after var4 var5 固定效应

var4 var5是控制变量，主要关注 var7*after的系数是否显著

静态面板实证方法

方法 4： 双重差分

- diff var6, t(var7) p(After) ,
- 运行上述命令，得到的是treat组和control组在事件前后的var6的均值以及对比

DIFFERENCE IN DIFFERENCES ESTIMATION

	BASE LINE			FOLLOW UP			
Outcome Variable	Control	Treated	Diff(BL)	Control	Treated	Diff(FU)	DIFF-IN-DIFF
var3	0.282	-0.759	-1.040	0.142	0.338	0.196	1.237
Std. Error	0.520	0.599	0.794	0.551	0.409	0.686	1.049
t	0.54	-1.27	-1.31	0.26	0.83	0.29	1.18
P>|t|	0.588	0.206	0.190	0.797	0.408	0.775	0.239

* Means and Standard Errors are estimated by linear regression
**Inference: *** p<0.01; ** p<0.05; * p<0.1

- Baseline指的是事件发生前，follow up是事件发生后
- Diff-in-diff的值1.237=(0.338+0.759)-(0.142-0.282)
- 可以根据不同要求在命令后面加不同的option

静态面板实证方法

方法4：双重差分

- 举例
- Compulsory Licensing: Evidence from the Trading with the Enemy Act. Petra Moser and Alessandra Voena, AER(2012)
- The Trading with the enemy Act(TWEA)是美国联邦政府在1917年通过的法律，赋予总统限制美国企业与战争中的敌国进行贸易的权利
- TWEA颁布以后，美国的外资管理部门宣布美国的企业可以不通过敌国的允许而使用他们的专利，这种行为称为强制性许可。在TWEA之下，美国一共没收了4500个化学行业的发明，其中727个专利被许可给326家企业（1919-1926）
- 文章研究这种强制性许可对美国国内的创新活动的影响，是促进还是抑制？

静态面板实证方法

方法4：双重差分

- 样本选择：1875-1945，化学染料行业
- 因变量(Y)：每一年每个子行业美国国内投资者的发明数量（即创新）
- 分组变量(Treat): 至少获得一项专利许可的子行业Treat取1，否则取0
- 时间变量(postTWEA): 即前面提到的after, TWEA颁布之后，即1918年及以后After取1，否则取0

$$Patents\ by\ US\ inventors_{c,t} = \alpha_0 + \beta' \mathbf{TREAT_c} \cdot postTWEA_t + \gamma \cdot Z_{c,t} + \delta_t + f_c + \varepsilon_{c,t},$$

没有加入treat和*post*TWEA单独项是因为模型控制了公司个体效应

静态面板实证方法

方法 4： 双重差分

	(1)	(2)
Subclass has at least one license	0.151***	0.255***
	(0.036)	(0.038)
Number of licenses		
Number of licenses squared		
Remaining lifetime of licensed patents		
Remaining lifetime of licensed patents squared (× 100)		
Number of patents by foreign inventors	0.283***	
	(0.018)	
Subclass fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Observations	471,120	471,120
Number of subclasses	7,248	7,248

treat * *post*TWEA

- 强制性许可有利于美国国内发明创新的进步

静态面板实证方法

方法 4： 双重差分

- 平行趋势检验方法：

  （1）比较两组样本事前的增长率是否有显著差异；

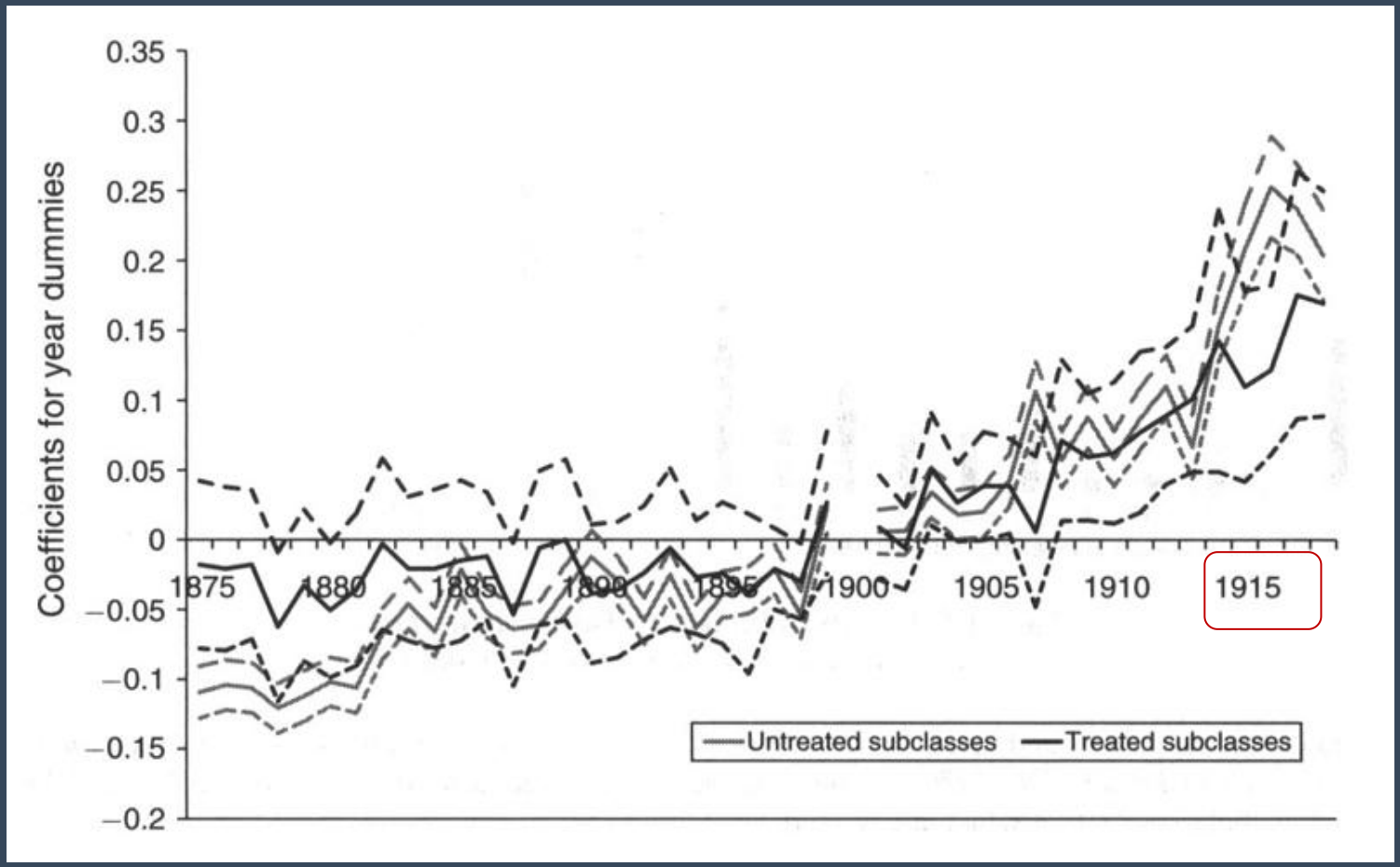
  （2）允许在事件发生前β_3随时间改变，加入事件前各年份的虚拟变量与处理组的交互项，如果不显著，说明平行趋势假设满足。如果不满足平行趋势假设，考虑其他因素加入控制变量。

$$Patents\ by\ US\ inventors_{c,t} = \alpha_0 + \beta_t \cdot YEAR_t \cdot TREAT_c \cdot pre1919_t + \gamma \cdot Z_{c,t} + \delta_t + f_c + \varepsilon_{c,t}.$$

- YEAR 年度 =1875、1876、1877、1878……1945,回归得出β系数，然后得到1875年至-1918年的β值

静态面板实证方法

方法 4： 双重差分



静态面板实证方法

方法4：双重差分



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Does banking competition affect innovation?☆

Jess Cornaggia[a], Yifei Mao[b], Xuan Tian[b,c,*], Brian Wolfe[d]

[a] McDonough School of Business, Georgetown University, USA
[b] Kelley School of Business, Indiana University, USA
[c] PBC School of Finance, Tsinghua University, China
[d] School of Management, State University of New York at Buffalo, USA

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ABSTRACT

We exploit the deregulation of interstate bank branching laws to test whether banking competition affects innovation. We find robust evidence that banking competition reduces state-level innovation by public corporations headquartered within deregulating states. Innovation increases among private firms that are dependent on external finance and that have limited access to credit from local banks. We argue that banking competition enables small, innovative firms to secure financing instead of being acquired by public corporations. Therefore, banking competition reduces the supply of innovative targets, which reduces the portion of state-level innovation attributable to public corporations. Overall, these results shed light on the real effects of banking competition and the determinants of innovation.

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- 银行竞争与企业创新之间的关系，看似非常明确，首先竞争带来融资，尤其是对小企业融资难度下降。根据熊彼特理论，外部贷款有助于创新。同时众多的文献也证明了银行业竞争能够促进企业创新，但Does Banking Competition Affect Innovation?却表明：在州际层面上，银行业竞争会降低企业的整体创新水平，其中创新水平以专利度量。

静态面板实证方法

方法 4： 双重差分

- 经验结果发现银行竞争减少了上市公司的创新，而提升了私营企业的创新。
- 第一，外部融资依赖性的不同。银行业竞争缓解了外部融资依赖性强的私营企业的融资约束，由于融资渠道的限制，更多依赖外部融资，因此对银行借贷融资以及当地银行业情况的变化非常敏感。而上市公司业绩稳定，通过资本市场来融资，因此其对于银行借贷融资的敏感度较低
- 第二，银行业放松管制扩展了企业融资渠道，并打破先前的银企关系，之前的私营企业由于难以打破当地银行与上市公司稳定的关系，只能从州外获得融资。放松管制事件后，私营企业的融资渠道拓宽，并获得对本公司创新计划的融资；
- 第三，银行业竞争使得私营企业得以存活，保持其独立性，并避免了被上市公司的收购。上市公司对创新性企业的收购，能够增加上市公司的创新，但银行竞争阻碍了这一渠道。

静态面板实证方法

方法4：双重差分

- 内生性问题。利用《1994年里格尔-尼尔州际银行和分行效率法》中，前后不同州执行这一法案的时间与程度的差异作为外生冲击。该法案允许跨州和在本州设立分支机构，这一法案被认为具有“分水岭”的意义。尽管银行业放松管制法提供了外生变化，但由于每个州可能自主选择放松管制的时间和方式，因而会引入第一个内生性问题，即反向因果。使用的双重差分方法（思想）。

$$LnPat_{i,t}(LnCite_{i,t}) = \alpha + \beta_1 Before_{i,e}^{2+} + \beta_2 Before_{i,t}^{1} + \beta_3 After_{i,t}^{1} + \beta_4 After_{i,t}^{2+} + Year_t + State_i + \varepsilon_{i,t}, \quad (2)$$

- 将银行业竞争指数设置为4个虚拟变量：放松管制的2年前，放松管制的前1年，放松管制的后1年，放松管制的2年后

静态面板实证方法

方法4：双重差分

Table 3
Endogeneity tests.

This table reports OLS regression estimates of Eq. (2). We truncate the state-year panel to a ± ten-year window around deregulatory events. Definitions of variables are in Appendix A. Robust standard errors clustered by year are reported in parentheses. Significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively.

	LnPat			*LnCite*		
	Total (1)	Public (2)	Private (3)	Total (4)	Public (5)	Private (6)
$Before^{2+}$	0.033	0.060	−0.006	0.014	−0.026	0.066
	(0.058)	(0.037)	(0.052)	(0.035)	(0.073)	(0.047)
$Before^{1}$	−0.103	−0.031	−0.111	−0.067	−0.094	−0.019
	(0.095)	(0.032)	(0.089)	(0.057)	(0.131)	(0.073)
$After^{1}$	−0.086*	−0.081**	−0.063	−0.028	−0.056	0.001
	(0.050)	(0.030)	(0.049)	(0.049)	(0.124)	(0.036)
$After^{2+}$	−0.130*	−0.091***	−0.100	−0.075**	−0.169*	−0.045
	(0.070)	(0.029)	(0.065)	(0.031)	(0.083)	(0.040)
Constant	0.627***	0.573***	0.012	2.351***	2.432***	−0.132
	(0.117)	(0.075)	(0.105)	(0.070)	(0.146)	(0.095)
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	910	910	910	910	910	910
Adjusted R^2	0.936	0.925	0.924	0.940	0.893	0.937

静态面板实证方法

方法4：双重差分

The Sensitivity of Corporate Cash Holdings to Corporate Governance

Qi Chen
Fuqua School of Business, Duke University

Xiao Chen
School of Economics and Management, Tsinghua University

Katherine Schipper
Fuqua School of Business, Duke University

Yongxin Xu
School of Economics and Management, Tsinghua University

Jian Xue
School of Economics and Management, Tsinghua University

The average cash holdings of Chinese-listed firms decreased significantly after the split share structure reform in China, which specified a process that allowed previously nontradable shares held by controlling shareholders to be freely tradable on the exchanges. The reduction in cash holdings is greater for firms with weaker governance and firms facing more financial constraints prior to the reform. The reform also significantly reduced the average corporate savings rate, as measured by cash-to-cash-flow sensitivity. These findings are consistent with the premise that the reform removed a significant market friction, which led to better incentive alignment between controlling shareholders and minority shareholders and relaxed financial constraints. Additional analyses show that the reform affects firms' cash management policies, investment decisions, dividend payout policies, and financing choices differently in private firms than in state-owned enterprises. (*JEL* G32, G34, G35)

- 2005年的股权分置改革同等于公司治理机制的一个外生冲击。改革前，较弱的公司治理和融资约束程度重的公司，其现金持有量在改革后减少更多。改革减少了市场摩擦，导致控股股东与中小股东之间的激励协同而缓解了融资约束。

静态面板实证方法

方法4：双重差分

Table 2
Average effect of the reform on cash holdings

	(1)	(2)	(3)	(4)	(5)	(6)
Reform	−0.0274***	−0.0483***		−0.0286**	−0.0554***	
	[2.75]	[3.82]		[2.30]	[3.45]	
*Reform*SOE*		0.0317***	0.0340***		0.0405***	0.0412***
		[2.92]	[3.21]		[2.95]	[3.06]
LogSales	−0.0246***	−0.0249***	−0.0261***	−0.0232***	−0.0234***	−0.0245***
	[3.15]	[3.18]	[3.32]	[3.16]	[3.19]	[3.32]
Leverage	0.338***	0.336***	0.336***	0.334***	0.332***	0.333***
	[10.84]	[10.77]	[10.77]	[10.69]	[10.63]	[10.65]
OPCF	0.492***	0.493***	0.495***	0.483***	0.484***	0.484***
	[16.01]	[15.99]	[16.08]	[15.35]	[15.35]	[15.41]
NETWC	−0.0103	−0.00857	−0.00834	−0.0167	−0.0160	−0.0157
	[0.41]	[0.34]	[0.33]	[0.67]	[0.64]	[0.63]
CashVol	0.574***	0.606***	0.616***	0.542***	0.559***	0.565***
	[2.94]	[3.10]	[3.14]	[2.73]	[2.81]	[2.83]
CAPEX	−0.0609*	−0.0670*	−0.0677*	−0.0806**	−0.0845**	−0.0852**
	[1.66]	[1.84]	[1.86]	[2.14]	[2.25]	[2.26]
Age	0.00250	0.00219	−0.00236	0.00298	0.00258	−0.00262
	[1.20]	[1.06]	[1.16]	[1.42]	[1.25]	[1.23]
M2B	0.00154	0.00142	0.00167	0.00654***	0.00649***	0.00650***
	[0.73]	[0.67]	[0.79]	[2.93]	[2.91]	[2.91]
IPO	0.213***	0.212***	0.211***	0.221***	0.220***	0.221***
	[21.25]	[21.25]	[21.15]	[20.01]	[20.00]	[20.02]
Dividend	0.0340***	0.0345***	0.0345***	0.0332***	0.0335***	0.0334***
	[7.83]	[7.95]	[7.96]	[7.60]	[7.68]	[7.64]
SEO	0.0940***	0.0942***	0.0938***	0.0911***	0.0912***	0.0911***
	[11.26]	[11.27]	[11.24]	[10.82]	[10.83]	[10.83]
Dum_ST	−0.0218	−0.0223	−0.0206	−0.0150	−0.0146	−0.0135
	[1.48]	[1.51]	[1.39]	[0.95]	[0.93]	[0.85]
SOE	−0.00276	−0.0165	−0.0183*	−0.000853	−0.0166	−0.0175
	[0.26]	[1.49]	[1.65]	[0.08]	[1.48]	[1.56]
$Before^2$			0.0106			0.0143
			[1.11]			[1.46]
$Before^1$			0.0122			0.0164
			[1.01]			[1.28]
$After^0$			−0.0366***			−0.0334**
			[2.68]			[2.23]
$After^1$			−0.0320**			−0.0265*
			[2.51]			[1.91]
$After^2$			−0.0108			−0.0132
			[1.16]			[1.42]
*Reform*TimingFactors*$_{04}$						
included	No	No	No	Yes	Yes	Yes
N	8,898	8,898	8,898	8,292	8,292	8,292

静态面板实证方法

方法4：双重差分

The Stock Market and Bank Risk-Taking
Antonio Falato and David Scharfstein
NBER Working Paper No. 22689
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JEL No. G01,G2,G21

ABSTRACT

We present evidence that pressure to maximize short-term stock prices and earnings leads banks to increase risk. We start by showing that banks increase risk when they transition from private to public ownership through a public listing or an acquisition. The increase in risk is greater than for a control group of banks that intended but failed to transition from private to public ownership, a result that is robust to using a plausibly exogenous instrument for failed transitions. The increase in risk is also greater than for a control group of banks that were acquired but did not change their listing status. We establish that pressure to maximize short-term stock prices helps to explain these findings by showing that the increase in risk is larger for newly public banks that are more focused on short-term stock prices and performance.

静态面板实证方法

方法4：双重差分

More formally, to examine the effect of private-to-public transitions on bank risk-taking, we use the following baseline DD regression specification:

$$RISK_{it} = \alpha + \beta_1 \times After_{it} + \beta_2 \times After_{it} \times Treatment_i + \gamma \times Z_{it} + \mu_t + \mu_i + \varepsilon_{it} \qquad (1)$$

where *i* and *t* index commercial banks and year-quarters. *RISK* is measured by the variety of supervisory ratings and financial information. *After* is an indicator variable that takes a value of

每个个体在时间上有差异，
事件发生的时间不同

静态面板实证方法

方法4：双重差分

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Does Stock Liquidity Enhance or Impede Firm Innovation?

VIVIAN W. FANG, XUAN TIAN, and SHERI TICE*

ABSTRACT

We aim to tackle the longstanding debate on whether stock liquidity enhances or impedes firm innovation. This topic is of interest because innovation is crucial for firm- and national-level competitiveness and stock liquidity can be altered by financial market regulations. Using a difference-in-differences approach that relies on the exogenous variation in liquidity generated by regulatory changes, we find that an increase in liquidity causes a reduction in future innovation. We identify two possible mechanisms through which liquidity impedes innovation: increased exposure to hostile takeovers and higher presence of institutional investors who do not actively gather information or monitor.

- 股票流动性的增加会导致未来创新的减少。
- 两种机制：股票流动性高，增加了敌意收购的风险，促使公司创新意愿不足；更多机构投资者不积极收集信息或进行监督。

静态面板实证方法

方法4：双重差分

$$INNOV_PAT_{i,t+n}(INNOV_CITE_{i,t+n}) = a + bILLIQ_{i,t} + c'CONTROLS_{i,t} + YR_t + FIRM_i + error_{i,t}, \quad (1)$$

- i为公司，t为时间，n等于1、2、3。因变量创新：1加上申请和最终批准的专利数量的自然对数和1加上每个专利的非自引次数的自然对数。
- 流动性测度，即相对有效价差(relative effective spread, ILLIQ)，为每年相对有效价差的自然对数。RESPRD的定义为执行价格与现行买卖报价中点之差的绝对值，除以现行买卖报价中点

静态面板实证方法

方法4：双重差分

Panel A: Innovation Measured by *INNOV_PAT*

Dependent Variable	(1) $INNOV_PAT_{t+1}$	(2) $INNOV_PAT_{t+2}$	(3) $INNOV_PAT_{t+3}$
$ILLIQ_t$	0.141***	0.168***	0.170***
	(0.020)	(0.023)	(0.026)
LN_MV_t	0.160***	0.090***	0.021
	(0.018)	(0.019)	(0.021)
$RDTA_t$	0.283***	0.265***	0.183*
	(0.089)	(0.095)	(0.095)
ROA_t	−0.032	0.247***	0.404***
	(0.068)	(0.068)	(0.082)
$PPETA_t$	0.287***	0.357***	0.481***
	(0.094)	(0.109)	(0.131)
LEV_t	−0.256***	−0.366***	−0.475***
	(0.075)	(0.084)	(0.092)
$CAPEXTA_t$	0.175	0.355***	0.175
	(0.119)	(0.134)	(0.148)
$HINDEX_t$	0.106	0.109	0.127
	(0.086)	(0.099)	(0.108)
$HINDEX^2{}_t$	−0.112	−0.072	−0.187
	(0.150)	(0.167)	(0.180)
Q_t	−0.006	0.011	0.005
	(0.007)	(0.008)	(0.009)
$KZINDEX_t$	−0.000*	−0.000	0.000
	(0.000)	(0.000)	(0.000)
LN_AGE_t	0.168***	0.190***	0.216***
	(0.035)	(0.038)	(0.042)
INTERCEPT	0.271**	0.757***	1.078***
	(0.106)	(0.116)	(0.127)
Year and firm fixed effects	Included	Included	Included
Number of obs. used	39,469	33,098	27,363
Adjusted R^2	0.839	0.844	0.849

Panel B: Innovation Measured by *INNOV_CITE*

Dependent Variable	(1) $INNOV_CITE_{t+1}$	(2) $INNOV_CITE_{t+2}$	(3) $INNOV_CITE_{t+3}$
$ILLIQ_t$	0.104***	0.106***	0.106***
	(0.015)	(0.016)	(0.019)
LN_MV_t	0.060***	0.014	−0.017
	(0.013)	(0.014)	(0.016)

Table II—*Continued*

Panel B: Innovation Measured by *INNOV_CITE*

Dependent Variable	(1) $INNOV_CITE_{t+1}$	(2) $INNOV_CITE_{t+2}$	(3) $INNOV_CITE_{t+3}$
$RDTA_t$	0.169**	0.149*	0.175*
	(0.080)	(0.090)	(0.098)
ROA_t	0.137**	0.299***	0.250***
	(0.061)	(0.062)	(0.074)
$PPETA_t$	0.168**	0.143	0.164*
	(0.077)	(0.087)	(0.095)
LEV_t	−0.197***	−0.266***	−0.313***
	(0.052)	(0.060)	(0.064)
$CAPEXTA_t$	0.240**	0.229*	0.243*
	(0.113)	(0.120)	(0.126)
$HINDEX_t$	0.129*	0.105	0.088
	(0.077)	(0.082)	(0.086)
$HINDEX^2{}_t$	−0.167	−0.035	−0.082
	(0.126)	(0.132)	(0.136)
Q_t	0.004	0.008	0.008
	(0.006)	(0.006)	(0.006)
$KZINDEX_t$	−0.000	−0.000	0.000
	(0.000)	(0.000)	(0.000)
LN_AGE_t	0.091***	0.063**	0.084***
	(0.025)	(0.029)	(0.031)
INTERCEPT	0.661***	1.010***	1.119***
	(0.080)	(0.089)	(0.098)
Year and firm fixed effects	Included	Included	Included
number of obs. used	39,469	33,098	27,363
Adjusted R^2	0.652	0.653	0.653

静态面板实证方法

方法4：双重差分

十进制报价冲击

- 十进制报价：样本期间对股票流动性的一个巨大的外生冲击。
- 2000年之前，美国三大交易所的最小报价和交易单位为1/16美元。2000年8月28日至2001年1月29日期间，纽约证券交易所和美国证券交易所将最小报价单位降至美分，并终止了分数定价系统。
- 2001年3月12日至2001年4月9日期间，纳斯达克进行了十进制计价（0.01美元）。
- 十进制化导致流动性显著增加，特别是在活跃交易的股票中。十进制直接影响流动性，不太可能直接影响创新，因此研究十进制引起的流动性变化后创新的变化，提供了一个准自然的实验。

静态面板实证方法

方法4：双重差分

- 倾向得分匹配PSM
- **处理组和控制组**
- 首先测量从十进制前的年份（-1年）到十进制后的年份（+1年）的年度相对有效价差（ΔRESPRD）的变化，其中0年表示公司十进制发生的会计年度。
- 根据$\Delta RESPRD_{-1to+1}$将3375家样本公司分成三等分，只保留最高三分之一的样本，即在十进制化前后的相对有效价差下降幅度最大的1125家公司，和最低三分之一的样本，即相对有效价差下降幅度最小的1125家公司。
- 每个位于最高三分之一的公司(处理组公司)与位于最低三分之一的公司(控制组公司)倾向得分最接近的公司相匹配。最终得到508对唯一匹配的公司（多个匹配下保留倾向分数之间的距离最小的那一对公司）

静态面板实证方法

方法 4： 双重差分

- 首先根据顶部和底部区域中的2,250个样本公司估计probit模型。
- 因变量：如果公司-年变量属于处理组tercile(顶端)则为1，否则为0。

  包括来自等式（1）的所有控制变量，这些控制变量在十进制前一年进行度量；

  机构持股为公司i的机构持股在十进制前一年的自然对数；

  Fama-French12个行业虚拟变量；

  十进制化前的创新增长变量（例如十进制化前专利数量PAT GROWTH和专利非自引数量CITE GROWTH增长）

静态面板实证方法

方法4：双重差分

- 平行趋势
- 检验一。重新运行probit模型（仅限匹配的样本）。所有自变量均无统计学意义。对冲击前创新增长变量的系数估计在统计学上不显著，这表明两组企业的创新结果没有明显的不同趋势。

Panel B: Prematch Propensity Score Regression and Postmatch Diagnostic Regression

	Dummy = 1 if in treatment group; 0 if in control group	
Dependent Variable	(1) Prematch	(2) Postmatch
$ILLIQ_{-1}$	0.848***	0.043
	(0.072)	(0.100)
LN_MV_{-1}	0.178***	0.006
	(0.033)	(0.046)
$RDTA_{-1}$	−0.435	0.315
	(0.425)	(0.582)
ROA_{-1}	0.774***	−0.041
	(0.178)	(0.245)
$PPETA_{-1}$	0.871***	0.052
	(0.193)	(0.287)
LEV_{-1}	0.051	−0.024
	(0.154)	(0.217)
$CAPEXTA_{-1}$	−2.135***	−0.255
	(0.562)	(0.789)
$HINDEX_{-1}$	−0.118	−0.083
	(0.388)	(0.530)
$HINDEX^2{}_{-1}$	0.407	0.112
	(0.513)	(0.676)
Q_{-1}	0.049***	0.009
	(0.017)	(0.023)
$KZINDEX_{-1}$	0.002***	−0.001
	(0.001)	(0.001)
LN_AGE_{-1}	0.025	0.082
	(0.036)	(0.052)
LN_INST_{-1}	1.291***	−0.003
	(0.147)	(0.213)
$PAT_GROWTH_{-3\text{ to }-1}$	0.078	0.022
	(0.086)	(0.110)
$CITE_GROWTH_{-3\text{ to }-1}$	0.010	−0.015
	(0.076)	(0.097)
INTERCEPT	2.110***	0.140
	(0.501)	(0.634)
Industry Fixed Effects	Included	Included
Number of obs. used	2,250	1,016
p-value of χ^2	<0.001	0.985
Pseudo R^2	0.157	0.008

静态面板实证方法

方法4：双重差分　□ 平行趋势

- 检验二。检查处理组公司和控制组公司的倾向分数之间的差异。差别相当小。例如，两个匹配的公司的倾向评分之间的最大距离只有0.024，而第95百分位的距离只有0.001。

Panel C: Estimated Propensity Score Distributions

Propensity Scores	No. of Obs.	Min	P5	P50	Mean	*SD*	P95	Max
Treatment	508	0.028	0.181	0.477	0.478	0.189	0.614	0.938
Control	508	0.042	0.182	0.477	0.478	0.189	0.614	0.938
Difference	–	0.000	0.000	0.000	0.000	0.001	0.001	0.024

静态面板实证方法

方法4：双重差分　□ 平行趋势

- 检验三。单变量比较。 如图所示，在十进制前，处理组公司和对照公司的特征之间的差异均无统计学意义

Panel D: Differences in Predecimalization Characteristics

	Treatment	Control	Difference	*t*-statistic
$ILLIQ_{-1}$	−3.975	−4.000	0.025	0.479
LN_MV_{-1}	4.812	4.829	−0.017	−0.149
$RDTA_{-1}$	0.064	0.054	0.010	1.571
ROA_{-1}	0.032	0.044	−0.012	−0.888
$PPETA_{-1}$	0.239	0.245	−0.006	−0.458
LEV_{-1}	0.193	0.200	−0.007	−0.563
$CAPEXTA_{-1}$	0.059	0.062	−0.003	−0.622
$HINDEX_{-1}$	0.146	0.149	−0.003	−0.271
Q_{-1}	2.303	2.137	0.166	1.130
$KZINDEX_{-1}$	−13.454	−11.414	−2.040	−0.865
LN_AGE_{-1}	2.253	2.192	0.061	1.154
LN_INST_{-1}	0.284	0.287	−0.003	−0.231
$PAT_GROWTH_{-3\text{ to }-1}$	−0.033	−0.024	0.009	0.287
$CITE_GROWTH_{-3\text{ to }-1}$	−0.058	−0.044	−0.015	−0.416

静态面板实证方法

方法4：双重差分　□ 平行趋势

- 检验四。图1为以十进制年为中心的7年期间内处理和控制组的专利数量，图2类似，为每个专利的非自引数量。每个节点的垂线反映了平均值的两个标准误差。
- 在十进制化前三年，处理和控制组专利(引用)数量的两条线几乎平行。在十进制化之后，两条线开始下降收敛，表明两组的创新生产率都下降了，而处理组的下降幅度更大。

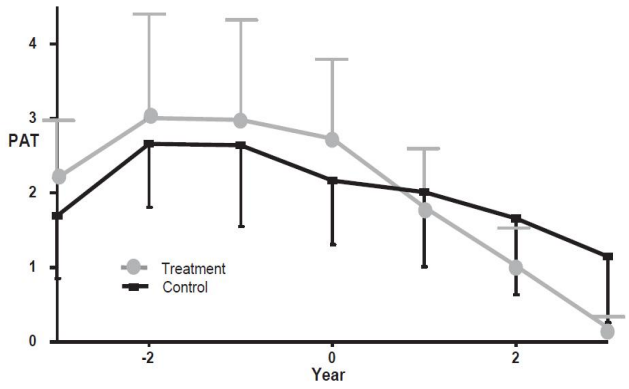


Figure 1. Number of patents surrounding decimalization. This figure shows the average innovation captured by the mean number of patents for treatment and control firms, from three years before decimalization to three years after decimalization. Decimalization year is denoted as year 0. The sample comprises 508 treatment firms and 508 unique control firms matched based on the procedures described in Table III. Two standard errors are represented by the vertical lines from each of the annual mean nodes.

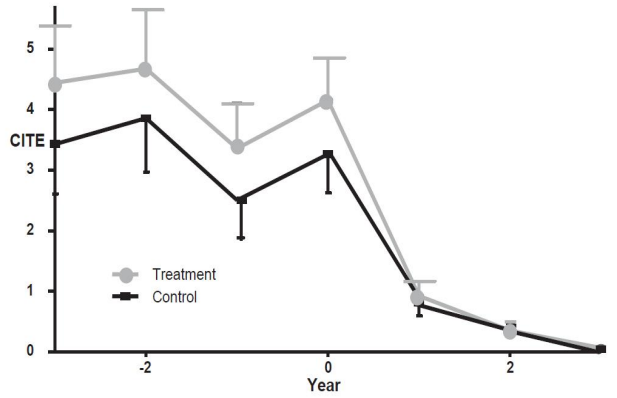


Figure 2. Citations per patent surrounding decimalization. This figure shows the average innovation captured by the mean number of citations per patent for treatment and control firms, from three years before decimalization to three years after decimalization. Decimalization year is denoted as year 0. The sample comprises 508 treatment firms and 508 unique control firms matched based on the procedures described in Table III. Two standard errors are represented by the vertical lines from each of the annual mean nodes.

静态面板实证方法

方法4：双重差分

首先，十进制后，处理组和控制组的创新生产率都下降，即流动性平均而言与公司创新呈负相关。其次，控制组的创新生产率下降幅度大于控制组，因为PAT和CITE上的DID估计值均为5%显著。

Panel E: Difference-in-Differences Test

	Mean Treatment Difference (after – before)	Mean Control Difference (after – before)	Mean DiD Estimator (treat – control)	*t*-statistic for DiD Estimator
PAT	−5.169	−1.682	−3.487**	−2.265
	(1.103)	(1.074)	(1.540)	
CITE	−11.14	−8.522	−2.616**	−1.976
	(0.986)	(0.884)	(1.324)	

静态面板实证方法

方法4：双重差分

- How Does Credit Supply Expansion Affect the Real Economy？Household Demand Channels（JF，2020）

used in Jayaratne and Strahan (1996) is

$$Y_{st} = \alpha_s + \gamma_t + \beta * DEREG_{st} + \epsilon_{st}, \quad (9)$$

where $DEREG_{st}$ takes a value of zero before a state deregulates and one afterward. This specification also includes state and year fixed effects. The estimated β from this specification reflects the immediate effect of deregulation on Y by comparing states that deregulate in year t with states that have not yet deregulated.[32]

The specifications in this study are not designed to capture the short-term effect of deregulation, but instead are designed to capture the higher loading on the longer economic cycle that comes from having a more deregulated banking sector. For example, consider two states, one that deregulated in 1980 and the other in 1984. Equation (9) estimates the effect of deregulation by comparing differences between the two states from 1980 to 1984, but treats both states equally after 1984 when the aggregate credit supply expansion accelerated to its peak. In contrast, the specifications in Section III.B give a higher deregulation score to the state that deregulated in 1980 with the presumption that a state that deregulated in 1980 (versus 1984) will have a stronger loading on the aggregate credit supply expansion that accelerated from 1984 to 1989.

- DEREG为0和1虚拟变量，为是否放松银行管制。然而DID分析的是短期影响，而对放松银行管制是否产生作用要看长期影响。在（9）式中，假设两个州A和B，A是1980年放松银行管制，而B是1984年放松银行管制，所以等式（9）假设1984年之后两个周放松银行管制的影响是相同的。

动态面板实证方法

广义矩估计（GMM）

The methodology of Generalized Method of Moments (GMM) for panel data analyses, proposed by Arellano and Bond (1991) and then further developed by Blundell and Bond (1998), is employed here to control for endogeneity in our estimations.[5] Consider the following model

$$Z_{i,t} = \gamma_1 EX_{i,t} + \gamma_2 EW_{i,t} + \nu_i + \eta_{i,t}, \quad i = 1,\ldots,N;\ t = 1,\ldots,T, \tag{12}$$

where Z is a given dependent variable, EX is a vector of strictly exogenous covariates; EW denotes a vector of predetermined covariates and endogenous covariates (predetermined variables are assumed to be correlated with past errors, while endogenous ones are assumed to be correlated with past and present errors); ν_i is the unobserved group-level effect, and η_{it} is the error term, with the assumption that ν_i and η_{it} are independent for each i over all t, and that there is no autocorrelation in the η_{it}.

Covariate:协变量，即独立变量（解释变量），不为实验者所操纵，但仍影响实验结果。

动态面板实证方法

广义矩估计（GMM）

First, in order to eliminate the unobservable group-specific effects, we difference Equation (12) and then it can be rewritten as

$$Z_{i,t} - Z_{i,t-1} = \gamma_1(EX_{i,t} - EX_{i,t-1}) + \gamma_2(EW_{i,t} - EW_{i,t-1}) + (\eta_{i,t} - \eta_{i,t-1}). \quad (13)$$

Second, instrumental-variable approaches are applied to deal with the endogeneity of explanatory variables in Equation (13), where the predetermined and endogenous variables in first differences are instrumented with appropriate lags of the specified variables in levels, while strictly exogenous regressors are first-differenced for use as instruments in the first-differenced equation.

However, the efficiency of this instrumental approach may be relatively weak, given the fact that lagged levels are often poor instruments for first differences. Therefore, Blundell and Bond (1998) propose the System-GMM approach, in which the first-differenced estimator (i.e., Equation (13)) is combined with the estimator in levels (i.e., Equation (12)) to form a more efficient "system estimator": for the first-differenced equation, the instruments are the same as that discussed above; for the levels equation, predetermined and endogenous variables in levels are instrumented with appropriate lags of their own first differences, while the strictly exogenous regressors can directly enter the instrument matrix for use in the levels equation.

动态面板实证方法

- 广义矩估计（GMM）
- 系统GMM方法，将包含变量水平的原估计方程与一阶差分后的方程纳入同一系统，通过增加工具变量来提高估计的准确性。其中，水平变量的滞后项是一阶差分方程的工具变量，而一阶差分变量的滞后项是水平方程的工具变量。
- 用Arellano-Bond检验的零假设是“误差项存在序列自相关”，检验结果要满足一阶残差存在序列自相关，而二阶残差不存在序列自相关。
- Sargan检验的零假设是“模型不存在过度识别，是有效的工具变量”，Hansen检验的零假设是“工具变量和残差不相关，是有效的工具变量”。

动态面板实证方法

- 广义矩估计（GMM）
  - 软件实现：GMM
  - xi:xtabond2 var4 l.var4 var5 var6 var7 i.var2 i.var3, gmm(var4 var5 ,lag(1 6) collapse) iv(i.var2 i.var3) twostep robust small
  - 加入“noleveleq”进去就是Diff-GMM，不加则是系统GMM。 twostep robust small（稳健标准误， 显示小样本t和F统计量）
  - gmm( )除了外生变量以外的变量都可以放进去（内生变量、前定变量等），IV（ ）放外生变量

Collapse：“退化”，避免instrument proliferation 工具过多

动态面板实证方法

广义矩估计（GMM）

```
Arellano-Bond test for AR(1) in first differences: z =  -0.81  Pr > z =  0.420
Arellano-Bond test for AR(2) in first differences: z =   0.79  Pr > z =  0.431
---------------------------------------------------------------------------
Sargan test of overid. restrictions: chi2(8)     =  53.89  Prob > chi2 =  0.000
  (Not robust, but not weakened by many instruments.)
Hansen test of overid. restrictions: chi2(8)     =  11.37  Prob > chi2 =  0.182
  (Robust, but weakened by many instruments.)

Difference-in-Hansen tests of exogeneity of instrument subsets:
  GMM instruments for levels
    Hansen test excluding group:     chi2(6)     =   9.13  Prob > chi2 =  0.166
    Difference (null H = exogenous): chi2(2)     =   2.24  Prob > chi2 =  0.327
```

- Sargan test和Hansen test 都是检验工具变量是否是弱工具变量。用Hansen test。
- Difference-in-Hansen test 工具变量外生的有效性。第二个为检查差分方程设置。所以看第一个prob值。

- The Sargan–Hansen test or Sargan's test is a statistical test used for testing over-identifying restrictions in a statistical model It was proposed by John Denis Sargan in 1958, and several variants were derived by him in 1975. Lars Peter Hansen re-worked through the derivations in 1982 and showed that it can be extended to general non-linear GMM in a time series context.

- Sargan, J. D. (1958). "The Estimation of Economic Relationships Using Instrumental Variables". *Econometrica*. **26** (3): 393–415.
- Sargan, J. D. (1975). "Testing for misspecification after estimating using instrumental variables". *Mimeo*. London School of Economics.
- Hansen, Lars Peter (1982). "Large Sample Properties of Generalized Method of Moments Estimators". *Econometrica*. **50** (4): 1029 – 1054.

动态面板实证方法

广义矩估计（GMM）

- **Endogeneity and the dynamics of internal corporate governance（JFE，2012）**

We use a well-developed dynamic panel generalized method of moments (GMM) estimator to alleviate endogeneity concerns in two aspects of corporate governance research: the effect of board structure on firm performance and the determinants of board structure. The estimator incorporates the dynamic nature of internal governance choices to provide valid and powerful instruments that address unobserved heterogeneity and simultaneity. We re-examine the relation between board structure and performance using the GMM estimator in a panel of 6,000 firms over a period from 1991 to 2003, and find no causal relation between board structure and current firm performance. We illustrate why other commonly used estimators that ignore the dynamic relationship between current governance and past firm performance may be biased. We discuss where it may be appropriate to consider the dynamic panel GMM estimator in corporate governance research, as well as caveats to its use.

动态面板实证方法

- 广义矩估计（GMM）

Table 7

The effect of board structure on current firm performance.

In this table, we report results from the estimation of the model:

$$y_{it} = \alpha_1 + \kappa_1 y_{it-2} + \kappa_2 y_{it-4} + \beta \mathbf{X}_{it} + \gamma \mathbf{Z}_{it} + \theta \mathbf{D}_{it} + \eta_i + \epsilon_{it}, \quad t = 1997, 1999, 2001, 2003.$$

y_{it} is return on assets (*ROA*) which is defined as operating income divided by assets. $\mathbf{X}_{it}$ includes board size (*LogBSIZE*), board independence (*INDEP*), and a dummy variable which is one if the CEO is the board chair (*CEO_CHAIR*). $\mathbf{Z}_{it}$ includes firm size (*LogMVE*), market-to-book ratio (*MTB*), standard deviation of stock returns (*RETSTD*), number of business segments (*LogSEGMENTS*), and leverage (*DEBT*). $\mathbf{D}_{it}$ includes firm age (*LogAGE*) and year dummies. The results are based on a sample of 6,034 firms and 20,003 firm years selected every other year (1991, 1993, 1995, 1997, 1999, 2001, and 2003). The board variable data come from the Compact Disclosure database. The firm characteristics come from CRSP and Compustat. *t*-statistics are reported in parentheses. For the static models, it is assumed that $\kappa_1 = \kappa_2 = 0$. All *t*-statistics are based on robust, firm-clustered standard errors. *a*,*b*,*c* represent significance at the 1%, 5% and 10% level, respectively. *AR*(1) and *AR*(2) are tests for first-order and second-order serial correlation in the first-differenced residuals, under the null of no serial correlation. The Hansen test of over-identification is under the null that all instruments are valid. The Diff-in-Hansen test of exogeneity is under the null that instruments used for the equations in levels are exogenous. The instruments used in the GMM estimation are: differenced equations: $y_{it-6}, y_{it-8}, \mathbf{X}_{it-6}, \mathbf{X}_{it-8}, \mathbf{Z}_{it-6}, \mathbf{Z}_{it-8}, \Delta \mathbf{D}_{it}$; level equations: $\Delta y_{it-4}, \Delta \mathbf{X}_{it-4}, \Delta \mathbf{Z}_{it-4}, \mathbf{D}_{it}$.

动态面板实证方法

- 广义矩估计（GMM）

Dependent variable (*ROA*)	Static model		Dynamic model	
	Pooled OLS	Fixed effects	Pooled OLS	System GMM
LogBSIZE	−0.0262[a] (−5.67)	−0.0261[a] (−4.32)	−0.0033 (−0.73)	0.0183 (0.43)
INDEP	−0.0266[a] (−3.56)	0.0202[b] (2.48)	0.0061 (0.82)	−0.0109 (−0.14)
CEO_CHAIR	0.0025 (1.09)	0.0003 (0.13)	0.0018 (0.83)	−0.0127 (−0.55)
LogMVE	0.0234[a] (27.97)	0.0429[a] (21.29)	0.0070[a] (6.89)	0.0160[a] (2.87)
MTB	−0.0025[a] (−2.91)	0.0014 (1.34)	0.0070[a] (3.36)	−0.0137[b] (−2.11)
RETSTD	−0.2047[a] (−11.69)	−0.0117 (−0.94)	−0.0832[a] (−4.44)	−0.5207[c] (−1.77)
LogSEGMENTS	−0.0087[a] (−4.47)	−0.0074[a] (−3.06)	−0.0012 (−0.76)	−0.0068 (−0.75)
LogAGE	0.0056[a] (4.26)	0.0008 (0.27)	−0.0003 (−0.20)	−0.0295[a] (−2.63)
DEBT	−0.0307[a] (−3.37)	−0.0625[a] (−3.99)	−0.0040 (−0.45)	−0.0345 (−0.52)
ROA (*t*−2)			0.4833[a] (24.40)	0.7590[a] (3.06)
ROA (*t*−4)			0.1054[a] (6.00)	−0.1212 (−0.22)
R^2	0.27	0.11	0.41	
AR(1) test (*p*-value)				(0.00)
AR(2) test (*p*-value)				(0.87)
Hansen test of over-identification (*p*-value)				(0.41)
Diff-in-Hansen tests of exogeneity (*p*-value)				(0.23)

动态面板实证方法

广义矩估计（GMM）

- **Endogeneity and the dynamics of internal corporate governance（JFE，2012）**

Based on the preceding discussion, we obtained the GMM results presented in Table 7, using the following code in Stata (Version 9):

```
xi: xtabond2 roa l.roa l2.roa logbsize indep
ceo_chair logmve mtb retstd logsegments
logage debt i.year, gmm(roa logbsize indep
ceo_chair logmve mtb
retstd logsegments debt, lag(3 4) collapse)
iv(i.year logage) twostep robust small
```

The Stata command incorporates our assumption that only firm age and the year dummies are exogenous. Since our data are sampled every two years, "lag(3 4)" invokes instruments from $t-6$ and $t-8$, respectively. We use the "collapse" to avoid instrument proliferation and obtain the instrument set specified in Eqs. (27) and (32).

动态面板实证方法

- 广义矩估计（GMM）

If we are willing to assume that we have a *strictly exogenous* variable (say, `z`), `xtabond2` allows us to partition our dependent variables (in the spirit of Hausman and Taylor, 1981) into endogenous and exogenous variables, using `'gmmstyle'` and `'ivstyle'` commands:

```
xtabond2 y l.y x1 x2, gmm(y x1 x2, lag(a.))
iv(z) <(options)>
```

安慰剂检验 Placebo Test

- 安慰剂检验本质上是反事实检验。在经济学里，往往是对一个政策的检验，假设如果这个政策不存在，是否对其有效果。那么这种假设检验就是placebo test。
- 安慰剂检验最为典型的例子：新研发药品的测试。实验者将药品的实验对象区分为实验组和控制组，实验组给予需要评估的药品，控制组给予“安慰剂”（无认为作用的药物），被实验者不知道自己所在组别。利用该方法检验药物的实际效果。

安慰剂检验 Placebo Test

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Does Going Public Affect Innovation?

SHAI BERNSTEIN*

ABSTRACT

This paper investigates the effects of going public on innovation by comparing the innovation activity of firms that go public with firms that withdraw their initial public offering (IPO) filing and remain private. NASDAQ fluctuations during the book-building phase are used as an instrument for IPO completion. Using patent-based metrics, I find that the quality of internal innovation declines following the IPO, and firms experience both an exodus of skilled inventors and a decline in the productivity of the remaining inventors. However, public firms attract new human capital and acquire external innovation. The analysis reveals that going public changes firms' strategies in pursuing innovation.

Theoretically, in frictionless financial markets, selling equities publicly should have no bearing on subsequent innovation activity. Under financial frictions, however, the transition to public equity markets improves firms' access to capital, which can lead to an increase in innovation activity, because such activity is particularly likely to be sensitive to financing constraints (Arrow (1962), Hall and Lerner (2010)). On the other hand, agency problems associated with the transition to public equity markets may undermine firm incentives to innovate (Berle and Means (1932), Jensen and Meckling (1976)).

□上市后公司外部融资能力提高，导致公司创新活动增加，但上市后的代理成本问题抑制公司创新

安慰剂检验 Placebo Test

- Does Going Public Affect Innovation？（JF,2015）。公司选择上市将影响公司的创新产出。文章使用上市申请期间的股价收益（NSQD）作为公司上市的工具变量，检验公司上市对其创新产出的影响。

$$IPO_i = \alpha_2 + \beta_2 NSDQ_i + \gamma_2 Y_i^{pre} + X_i'\delta_2 + \nu_k + \mu_t + \varepsilon_{2i},$$

$$Y_i^{post} = \alpha_3 + \beta_3 \widehat{IPO}_i + \gamma_3 Y_i^{pre} + X_i'\delta_3 + \nu_k + \mu_t + \varepsilon_{3i},$$

- 问题：如何检验公司申请上市之后，是产权性质变更而非其他渠道对公司创新产生影响？。如果是除企业上市之外其他渠道的影响，则使用申请期以外的股票收益与公司创新数据进行回归，依然可以看到显著的结果。

安慰剂检验 Placebo Test

- 安慰剂检验：使用公司上市申请期间外的股票价格。

- **Returns following IPO outcome**：IPO 成功或注销之后两个月内的股价收益。
- **Returns in year before IPO filing**: IPO提出申请之前的一年时间的股价收益。
- **Returns in year after IPO filing**: IPO提出申请之后的一年时间的股价收益。

Table IV
Placebo Test

This table reports a placebo test to assess the validity of the instrumental variable exclusion restriction. The dependent variable is the average scaled citations in the five years after the IPO filing. *Returns following IPO filing* are the two-month NASDAQ returns calculated from the IPO filing date. *Returns following IPO outcome* are the two-month NASDAQ returns calculated from either the date of the equity issuance or the date of the IPO filing withdrawal. When the date of IPO filing withdrawal is not available, 270 days subsequent to the last amendment of the IPO filing is assumed (Lerner (1994)). *Returns in year before IPO filing* are the two-month NASDAQ returns calculated from a year before the IPO filing. *Returns in year after IPO filing* are the two-month NASDAQ returns calculated from a year after the IPO filing. The variables included in the regressions are prefiling average scaled citations, prefiling number of patents, *VC Backed*, *Pioneer*, *Early Follower*, and the three-month NASDAQ returns before the IPO filing. All variables are defined in the Appendix. The model is estimated using OLS, and robust standard errors are reported in parentheses. *, **, and *** indicate that the coefficient is statistically significant at the 10%, 5%, and 1% level, respectively.

Dependent Variable	(1) Scaled Citations	(2) Scaled Citations	(3) Scaled Citations	(4) Scaled Citations	(5) Scaled Citations	(6) Scaled Citations	(7) Scaled Citations
Returns following IPO filing	−0.498**				−0.482**	−0.495**	−0.509**
	(0.239)				(0.237)	(0.237)	(0.241)
Returns following IPO outcome		0.207			0.162		
		(0.251)			(0.248)		
Returns in year before IPO filing			0.201			0.193	
			(0.254)			(0.252)	
Returns in year after IPO filing				0.006			0.037
				(0.096)			(0.094)
Observations	1,079	1,079	1,079	1,079	1,079	1,079	1,079
R^2	0.242	0.240	0.239	0.239	0.242	0.242	0.242
Filing year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes

安慰剂检验 Placebo Test

- Zhangkai Huang, Lixing Li, Guangrong Ma, and Lixin Colin Xu: Hayek, Local Information, and Commanding Heights: Decentralizing State-Owned Enterprises in China （AER,2017）
- 改革开放后市场竞争加剧和私营部门崛起，政府开始面临国有企业亏损的问题，为提高国企业绩，国企权力下放成为国企改革的重要选择，比如上世纪90年代的“抓大放小”，大型国企合并形成工业集团，小型国企私有化分权。中国的权力下放全面转移了现金流和控制权，国企的财务、劳动、工资、人员以及资产负债、税收等都由中央政府转向地方政府。
- 当政府关心国企效率改善时，国企与监管政府的物理距离越大，其权力下放的可能性越大，因为离政府越近，政府获取的国企信息越多，对国企的控制力也较强，反之，距离越远导致沟通成本越高，信息不对称程度加大，政府对国企管控就越低。同时，这种距离-权力下放的正向关联（即距离越大越容易下放）会随着企业绩效异质性加大而加大。

安慰剂检验 Placebo Test

- 哈耶克(1945)认为，地方信息是理解替代经济系统效率的关键，以及生产是否应该集中或分散。国有企业距离中央政府的距离越远，国有企业越有可能发生权利下放。当交易成本更高时，这种影响会更为明显

$$(1)\quad \Pr(Decen_{ijkt}=1)=\Pr(Distance_{ik}\delta+X_{ijk,lag}\beta+Z_{lag}\alpha+\theta_j+\rho_k+\gamma_t+\varepsilon_{ijkt}>0).$$

- **Decentralized**：是否下放（虚拟变量，若企业的隶属关系变为较低级别政府，则等于1，反之为0）。
- **Distance**： 公司与中央政府之间的物理距离（公司与政治中心的距离）

安慰剂检验 Placebo Test

TABLE 3—DETERMINATION OF DECENTRALIZATION

	Probit				Hazard	Multinomial logit	
	Whole sample	Central SOEs	Provincial SOEs	Municipal SOEs	Whole sample	Whole sample	
	Decentralized$_{(t)}$					*Decentralized*$_{(t)}$	*Restructured*$_{(t)}$
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A. Baseline results							
Distance$_{lag}$	0.0054	0.0047	0.0037	0.0044	0.3059	0.0053	0.0011
	(0.0007)	(0.0017)	(0.0008)	(0.0005)	(0.0434)	(0.0008)	(0.0010)
log(firm asset$_{lag}$)	−0.0029	−0.0049	−0.0030	−0.0017	−0.1454	−0.0025	−0.0109
	(0.0005)	(0.0009)	(0.0010)	(0.0009)	(0.0320)	(0.0006)	(0.0011)
ROS$_{lag}$	−0.0102	−0.0171	−0.0129	−0.0044	−0.5428	−0.0074	−0.0683
	(0.0015)	(0.0065)	(0.0029)	(0.0017)	(0.0801)	(0.0015)	(0.0065)
Firm importance$_{lag}$	−0.0466	−0.2243	−0.1461	−0.0248	−2.5841	−0.0399	−0.0716
	(0.0203)	(0.5033)	(0.1049)	(0.0152)	(1.1532)	(0.0187)	(0.0200)
Fully state-owned$_{lag}$	−0.0069	−0.0129	−0.0147	0.0001	−0.3387	−0.0059	−0.0264
	(0.0013)	(0.0051)	(0.0023)	(0.0013)	(0.0723)	(0.0013)	(0.0045)
GDP per capita$_{lag}$	0.0061	−0.0026	−0.0094	0.0620	0.3009	0.0045	0.0007
	(0.0044)	(0.0050)	(0.0331)	(0.0224)	(0.2421)	(0.0043)	(0.0154)

安慰剂检验 Placebo Test

- 问题: 北京即是政治中心，同时也是中国的经济中心。使用监管政府和国企之间的距离事实上也包含了经济因素。找个经济中心，非政治中心。
- 文章使用安慰剂距离（Placebo distance）检验基准结果。安慰性距离（对于国家级国企是其与上海的距离；省级国企则是其与省内非省会最大城市的距离，对于市级国企则是其与非市中心的最大县级市距离）。此处的距离为企业经营所在地与政府所在地距离。
- 定义如下：
  - 央企和上海之间的距离。
  - 省属国企与省内非省会最大城市的距离。
  - 市属国企与最大的非自治县级市的距离

安慰剂检验 Placebo Test

TABLE 4—DETERMINANTS OF DECENTRALIZATION: PLACEBO TEST AND THIRD FRONT CONSTRUCTION AS IV

	Placebo test		2SLS	
	Whole sample (1)	Excluding small placebo city (2)	First stage *Distance* (3)	Second stage $Decentralized_{(t)}$ (4)
$Placebo\ distance_{lag}$	0.0005	0.0002		
	(0.0009)	(0.0012)		
$Distance_{lag}$				0.0240
				(0.0124)
TFC			0.2416	
			(0.1018)	
Controls	YES	YES	YES	YES
Observations	69,785	56,742	69,785	69,785
Kleibergen-Paap *F*-statistic			5.627	
Anderson-Rubin *F*-statistic				3.83
(*p*-value)				(0.0504)
90 percent confidence interval				[0.0050, 0.0537]

似不相关回归 SUR

- Cash-Flow Sensitivities and the Allocation of Internal Cash flow（RFS,2014）
- 公司现金流用于投资、增加现金持有量、现金股利、减少债务、股票回购。

$$Inv_t + \Delta Cash_t + Div_t - \Delta D_t - \Delta E_t = CF_t, \quad (1)$$

- seeming unrelated regression（SUR，似不相关回归）estimates are equivalent to equation-by-equation OLS estimates if the same set of explanatory variables is included in each equation。被解释变量不同，而解释变量相同。

似不相关回归 SUR

- SUR与分开回归的区别： SUR考虑了不同方程误差项之间的相关性。

$$Inv_{it}=\alpha^{Inv}CF_{it}+\beta^{Inv}MB_{it-1}+\gamma^{Inv}Y_{it-1}+f_i+y_t+\varepsilon_{it}^{Inv}, \quad (2)$$

$$\Delta Cash_{it}=\alpha^{\Delta Cash}CF_{it}+\beta^{\Delta Cash}MB_{it-1}+\gamma^{\Delta Cash}Y_{it-1}+f_i+y_t+\varepsilon_{it}^{\Delta Cash}, \quad (3)$$

$$Div_{it}=\alpha^{Div}CF_{it}+\beta^{Div}MB_{it-1}+\gamma^{Div}Y_{it-1}+f_i+y_t+\varepsilon_{it}^{Div}, \quad (4)$$

$$\Delta D_{it}=\alpha^{\Delta D}CF_{it}+\beta^{\Delta D}MB_{it-1}+\gamma^{\Delta D}Y_{it-1}+f_i+y_t+\varepsilon_{it}^{\Delta D}, \quad (5)$$

$$\Delta E_{it}=\alpha^{\Delta E}CF_{it}+\beta^{\Delta E}MB_{it-1}+\gamma^{\Delta E}Y_{it-1}+f_i+y_t+\varepsilon_{it}^{\Delta E}. \quad (6)$$

似不相关回归 SUR

Table 6
Are financing decisions symmetric for investments versus disinvestments and for profits versus operating losses?

	Dependent variable				
	Change in Cash Holdings	Short-term Debt Issues	Long-term Debt Issues	Equity Issues	Share Repurchases
Intercept	0.0945	−0.2658*	−0.2270***	0.6812***	0.0939***
	(0.4590)	(0.0846)	(0.0039)	(0.0034)	(0.0006)
Investment in Net Working Assets	−0.0393	0.1879***	0.1411***	0.6304***	−0.0013
	(0.1692)	(0.0002)	(0.0082)	(0.0001)	(0.1782)
Increases in Net Fixed Assets	0.0144	0.1929***	0.5417***	0.2856***	0.0058**
	(0.6662)	(0.0001)	(0.0001)	(0.0001)	(0.0170)
Decreases in Net Fixed Assets	0.2142	−0.2434	0.2569***	1.1937***	−0.0070*
	(0.1282)	(0.3341)	(0.0028)	(0.0060)	(0.0962)
Positive Income Available to Common and Preferred	0.1954***	−0.1201***	−0.1413***	−0.5386***	0.0045***
	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0013)
Negative Income Available to Common and Preferred	−0.0192	−0.1389***	−0.0913***	−0.8034***	−0.0145***
	(0.4848)	(0.0001)	(0.0004)	(0.0001)	(0.0052)
Dividends	0.0914	0.3008*	0.1101	0.7576***	0.0771***
	(0.5106)	(0.0916)	(0.4374)	(0.0001)	(0.0001)
Difference between Increases and Decreases in NFA	−0.1998	0.4363	0.2848***	−0.9081**	0.0128**
	(0.1515)	(0.1198)	(0.0003)	(0.0340)	(0.0203)
Difference between Positive and Negative Income	0.2146***	0.0187	−0.0500*	0.2648***	0.0190***
	(0.0001)	(0.4696)	(0.0989)	(0.0001)	(0.0008)
Difference between Increases in NFA and Negative Income	−0.0048	0.0540	0.4504***	−0.5178***	−0.0087
	(0.9053)	(0.2524)	(0.0001)	(0.0001)	(0.1408)
Difference between Decreases in NFA and Positive Income	0.4096***	−0.3636	0.1156	0.6551*	−0.0025
	(0.0040)	(0.1412)	(0.1418)	(0.0951)	(0.5589)

We collect the data from the Compustat annual files between 1972 and 2005. We exclude utilities (SIC codes 4900–4949) and financial firms (SIC codes 6000–6999). We estimate separate coefficients for increases and for decreases in Net Fixed Assets and estimate separate coefficients for positive and for negative Income Available to Common and Preferred shareholders. As explanatory variables we also add four firm characteristics for year $t-1$: the natural logarithm of firm assets as a measure of firm size, the ratio of firm market value to book value of assets, the ratio of earning to assets, and the ratio of long-term debt to total assets. For ease of interpretation, we standardize these variables by subtracting their annual means and dividing by their annual standard deviation. In the case of long-term debt-to-assets, we standardize using the yearly industry-specific mean and standard deviation. As explanatory variables we further add lagged dependent variables. The table presents the average of the 34 coefficient estimates from annual regressions. The p-value from a two-tailed t-test is presented in parentheses under each coefficient. The model estimated is asset-weighted Seemingly Unrelated Regressions (SUR). We impose the following restrictions:

$$
\begin{aligned}
-\alpha_1+\alpha_2+\alpha_3+\alpha_4-\alpha_5 &= 0\\
-\beta_{11}+\beta_{21}+\beta_{31}+\beta_{41}-\beta_{51} &= 1\\
-\beta_{12}+\beta_{22}+\beta_{32}+\beta_{42}-\beta_{52} &= 1\\
-\beta_{13}+\beta_{23}+\beta_{33}+\beta_{43}-\beta_{53} &= 1\\
-\beta_{14}+\beta_{24}+\beta_{34}+\beta_{44}-\beta_{54} &= -1\\
-\beta_{15}+\beta_{25}+\beta_{35}+\beta_{45}-\beta_{55} &= -1\\
-\beta_{16}+\beta_{26}+\beta_{36}+\beta_{46}-\beta_{56} &= 1\\
-\gamma_1+\gamma_2+\gamma_3+\gamma_4-\gamma_5 &= 0
\end{aligned}
$$

The α coefficients are the intercepts. The first subscript of the β coefficients indicates the column (i.e., the dependent variable) while the second subscript indicates the row (i.e., the independent variable). The γ coefficients are coefficients of size, market-to-book assets, profitability, leverage, and the lagged dependent variables. For brevity, the γ coefficients are not reported. *, **, *** indicate significance at the 0.10, 0.05, and 0.01 levels from a two-tailed t-test.

似不相关回归 SUR

- Stata实现：SUR
- 三方程SUR： xtsur (y1 y2 y3 = x1 x2 x3 x4), onestep